



**BLOCK 4**  
**INDUSTRIAL ACCIDENTS AND ITS**  
**PREVENTION**

THE PEOPLE'S  
UNIVERSITY

**MIS – 021**  
**Safety Philosophy and**  
**Principles of Accident**  
**Prevention**

Block

**4****INDUSTRIAL ACCIDENTS AND ITS PREVENTION****UNIT 13****Introduction to Industrial Accident** 337**UNIT 14****Types of Accidents and Its Analysis** 361**UNIT 15****Cost of Accidents** 384**UNIT 16****Prevention of Accidents** 400

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## **BLOCK 4 INDUSTRIAL ACCIDENTS AND ITS PREVENTION**

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Accidents do not happen all of a sudden. They are caused by the mismanagement of the events causing to accident. It is conceived that accidents are unexpected or unplanned events, however it is seen that most of the time, it is not. Majority of accidents are the result of unsafe conditions and work practices that have been ignored or tolerated for weeks, months, or even years. According to the recent estimates, some 2.3 million workers die from work-related accidents and diseases; over 474 million people suffer from occupational diseases and non-fatal accidents; with the costs of these exceeding US\$2.8 trillion, or 4% of gross domestic product. It is therefore necessary to follow the best safety practices along with the proper inspections to prevent industrial accidents.

This Block consists of four units.

The unit 13 entitled, 'Introduction to Industrial Accident' provides the definition of an accidents and gives a brief introduction to various types of accidents. It also provides details of various causes of industrial accidents and well defined methods for computation of various values as significant in accident terminology. The later section of the chapter talks in detail about the industrial accident in Indian scenario. Some of the worst industrial disasters in India are discussed in short in this section. Models of accident causation have also been discussed in this chapter. This chapter thus provides basic overview of the introduction to industrial accidents.

The unit 14 entitled, 'Types of Accidents and its Analysis' deals with the key factors involved in accident analysis. It discusses the simple accident investigation techniques that do not require the users to be a safety professional. Also, it talks about some of the advanced techniques too which are complex but systematic accident investigation techniques. This unit also provides brief description of techniques which helps in assessment of the safety of a system.

The unit 15 entitled, 'Cost of Accidents' deals with various costs involved in the event of industrial accident. Being a safety issue, accident is always undesirable. However, accident also has financial impact. Some of the examples from past accident show how much an accident can actually cost to a company. This unit thus deals with economic aspects of accident. It shows the different types of cost involved in an accident. This unit also provides a brief detail of different tools for accident cot analysis.

The unit 16 entitled, 'Prevention of Accidents' talks about the need for accident prevention and the ways to do the same. It is observed that many of the industrial accidents are actually preventable and if not preventable at all then at least the risk can be reduced to a considerable extent. This unit discusses the humanitarian factors associated in accidents and its prevention. Later section also provides details of prerequisites for hazard control system. The unit also suggests effective workplace inspections for accident prevention and discusses type of information needed to complete an effective inspection report. It also suggested some common practices to prevent accidents in the workplace.

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## UNIT 13 INTRODUCTION TO INDUSTRIAL ACCIDENT

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### Structure

- 13.1 Introduction
  - Objectives
- 13.2 Types of Accidents
- 13.3 Causes of Industrial Accidents
- 13.4 Important Terminologies
- 13.5 Indian Standard for Measurement of Industrial Accidents
- 13.6 Computation of Frequency, Severity and Incident Rate
- 13.7 Industrial Accident and Indian Scenario
- 13.8 Basic Steps Followed in Accident Investigation
- 13.9 Elements of Incident Investigation Forms
- 13.10 Models of Accident Causation
- 13.11 Illustrative Problem
- 13.12 Let Us Sum Up
- 13.13 Key Words
- 13.14 Answers to SAQs
- 13.15 References and Further Studies

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### 13.1 INTRODUCTION

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As per Oxford dictionary, “accident is an unfortunate incident that happens unexpectedly and unintentionally, typically resulting in damage or injury.”

Accidents do not happen all of a sudden, they are caused by the mismanagement of the events causing to accident. It is not simply because of human or technological failure, but because of the failure of Management Control Systems. Accidents are due to uncontrolled events or activities. It is a management function to control all events/activities in its physical, technological and human aspects. Recent studies suggest that around 2.3 million laborer mortalities occur due to work-related accidents and diseases. More than 474 million people endure chronic ailments, resulting from occupational activity, and non-fatal accidents. The costs of these accidents and/or ailment medication are of the order of US\$2.8 trillion or 4% of gross domestic product. These expenses are at least 10% more compared to its Figure in the previous decade, and can be anticipated to grow more due to rapid advancements in technology, changing nature of accidents, hazards and risks, changing societal views of accidents, introduction of new forms of regulations, and increasing levels of complexity and coupling.

## Objectives

After going through this unit, you should be able to

- discuss the causes of the accidents
- describe the Indian standard for measurement of industrial accidents
- estimate the computation of frequency, severity and incident Rate
- appraise the industrial accident and Indian scenario
- compare the elements of incident investigation forms discuss the models of accident causation

It is conceived that accidents are unexpected or unplanned events; however, it is observed that most of the time, it is not. Majority of accidents are the result of insecure work environments and practices that were overlooked or tolerated for long, *i.e.*, weeks, months, or years. Studies suggest that hazardous working conditions contribute a meagre 3 % among all other type of workplace accidents. Negligent attitude (a.k.a unsafe act) of the worker account for 95 % of all workplace accidents. Uncontrollable acts add 2 % of all workplace accidents. Thus, 98 % of all workplace accidents are controllable by the management.

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## 13.2 TYPES OF ACCIDENTS

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The different types of industrial accidents can be broadly classified as follows:

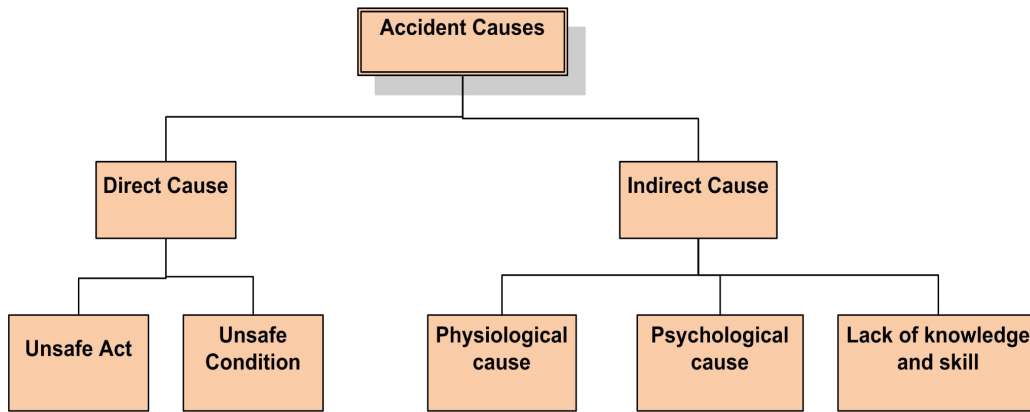
- 1) **Accident with direct consequence:** An unexpected and undesired event that is actuated unintentionally and apparently by chance. The results of the unfavorable events can be observed within a short duration. For instance, an accident where an operator is crushed in a press, an explosion, and the breakdown of an installation.
- 2) **Accident giving increased probability for injury or damage:** This is also unexpected and undesired event causing harm to a person, but its consequence(s) is(are) indirect. For example, there is the likelihood of lung problems due to exposure of a person to toxic gases of a chemical plant in case of any leakage/accident.
- 3) **Slow deterioration or degeneration:** Examples include occupational diseases or environmental destruction caused by continuous exposure or the absorption of repeated small doses of chemical substances, prolonged overexertion, etc.
- 4) **Sabotage:** A negative event caused by willful action of a person. Sometimes, this type of event is not categorized as an accident.

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## 13.3 CAUSES OF INDUSTRIAL ACCIDENTS

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Accidents do not happen all of a sudden. There are several causes which leads to the accidents. A hierarchical diagram presented below for the different causes of industrial accidents.



**Figure 13.1: Causes of Industrial Accidents**

**Unsafe act-** Unsafe act is one of the direct cause of the accident in industry and is directly related to the act of worker. Some common example includes;

- Working without safety devices
- Operating a machine with low/zero knowledge
- Operating the machine at unsafe speed
- Using incorrect tools and equipment for machine operation
- Unsafe loading/unloading practice
- Unsafe posture/position
- Working on moving equipment

Unsafe conditions include:

- Unguarded/inadequately guarded machines
- Use of defective machines, tools, equipment etc.
- Unsafe storing methods particularly for chemicals and gases
- Poor ventilation leading to suffocation
- Poor lighting conditions or illumination
- Poor house keeping
- Unsafe design of machine, equipment or work place.

#### **SAQ 1**

- a) What is an accident?
- b) What are the different types of accidents?
- c) What are the causes of industrial accident?

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## 13.4 IMPORTANT TERMINOLOGIES

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**Accident:** An accident can be described as an unplanned event or action that results in undesired consequences, *e.g.*, injury, ill health, damage to the environment, damage to or loss of property, plant and materials.

**Incident:** An incident is the sequence of events or actions that causes an accident. All accidents are subset of incidents. Incident is a superset which also includes dangerous occurrences and near misses.

A **near miss** is an unplanned event that did not result in injury, illness, or damage-but had the potential to do so. Only a fortunate break in the chain of events prevented an injury, fatality, or damage.

**Fatal and Non-fatal accident:** A fatal occupational injury results in loss of life, whereas a nonfatal occupational injury results in at least 4 days' absence from work.

**Undesired circumstance:** A set of conditions or circumstances that have the potential to cause injury or ill health, *e.g.*, untrained nurses handling heavy patients.

Incident rates are an indication of number of incidents and their severity. They are measurements *only* of past performance or lagging indicators.

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## 13.5 INDIAN STANDARD FOR MEASUREMENT OF INDUSTRIAL ACCIDENTS

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The Indian Standard, IS: 3786 -1983, titled "Methods for computation of Frequency and Severity Rates in Industrial Injuries and classification of industrial accidents" details step for recording and categorizing industrial accidents. Method for computation of frequency, severity and incident rate of work injuries in industrial premises has been provided in this and it also includes details of work-related injury.

Rules for calculation of man-days lost:

- a) Man-days lost due to temporary total disability;
- b) Man-days lost according to schedule of charges for death and permanent disabilities as given in table below. In case of multiple injury, the sum of schedule charges shall not be taken to exceed 6000 man-days;
- c) Days lost due to injury in previous periods, that is, if any accident which occurred in previous period is still causing loss of time in the period under review, such loss of time is also to be included in the period under review;
- d) In the case of intermittent loss of time, each period should be included in the severity rate for the period in which the time is lost; and

- e) If any injury is treated as a lost time injury in one statistical period and subsequently turns out to be a permanent disability; the man-days charged to the injury shall be subtracted from the schedule charge for the injury when permanent disability becomes known.

### Days of Disablement (Lost Time)

Charges for disabilities scheduled as per IS:3788-1983 norms for some injuries are given in table 13.1. In the case of death or disablement of a permanent nature whether it be partial or total disablement man-days lost means the charges in days of earning capacity lost due to such permanent disability or death. In other cases, the day on which the injury occurred or the day the injured person returned to work are not to be included as man-days lost; but all intervening calendar days (including Sundays or, days off, or days of plant shut down) are to be included. If after resumption of work, the person injured is again disabled for any period arising out of the injury which caused his earlier disablement, the period of such subsequent disablement is also to be included in the man-days lost.

### Man-Hours Worked

The total number of employee-hours worked by all employees working in the industrial premises. It includes managerial, supervisory, professional, technical, clerical and other workers including contractor's labor.

### Calculation of Man-Hours Worked

Man-hours worked shall be calculated from the pay roll or time clock recorded including overtime. When this is not feasible, the same shall be estimated by multiplying the total man-days worked for the period covered by the number of hours worked per day. The total number of man-days for a period is the sum of the number of men at work on each day of the period. If the daily hours vary from department to department separate estimates shall be made for each department and the result added together. There may be cases when actual man-hours are not used, in those cases that the basis on which the estimates are made should be properly indicated.

**Table 13.1: Charges for disabilities scheduled as per IS: 3788 – 1983 norms for some injuries**

| Injury description                                                                                                                                                                             | Loss of earning capacity in percentage | Equivalent man days lost |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------|
| Total Disablement/ Death<br>(includes total deafness, loss of vision, loss of both hands, loss of a hand and a foot, very severe facial disfigurement, Double amputation through leg of thigh) | 100                                    | 6000                     |

|                                                        |    |      |
|--------------------------------------------------------|----|------|
| Amputation (loss of limb) through shoulder joint       | 90 | 5400 |
| Loss of thumb                                          | 30 | 1800 |
| Loss of four fingers                                   | 50 | 3000 |
| Loss of three fingers                                  | 30 | 1800 |
| Loss of two fingers                                    | 20 | 1200 |
| Amputation at hip                                      | 90 | 5400 |
| Amputation of one foot resulting in end-bearing        | 30 | 1800 |
| One eye lost, without complications other being normal | 40 | 2400 |

**Refer IS: 3788 – 1983 for complete list**

**SAQ 2**

- What is the difference between accident, incident and a near miss?
- What is fatal and non-fatal accident?
- What is the Indian standard for measurement of industrial accident?
- What do you mean by manhours worked?

### **13.6 COMPUTATION OF FREQUENCY, SEVERITY AND INCIDENT RATE**

Several metrics, viz., Frequency Rate, Severity Rate and Incidence Rate, have been defined to measure/assess injuries. The ratio of the total number of injuries over corresponding number of man-hours worked is defined as Frequency Rate. The Severity Rate, which takes into account the duration of disability as well, is the ratio of man-hours lost due to injuries and the number of man-hours worked. Note that severity rate is evaluated only for those non-fatal injuries in which workers returned to their jobs during the same year. Similarly, the incidence rate is the ratio of the number of injuries by average daily employment per thousand. The above metrics along with their evaluation formula are detailed below.

**Frequency Rate:**

There are two different types of frequency rates, one for lost time injury and another for reportable lost time injury and both are required to be calculated.

$$FR_A = \frac{\text{Number of lost time injury} \times 10,00,000}{\text{Man – hours worked}}$$

$$FR_B = \frac{\text{Number of reportable lost time injury} \times 10,00,000}{\text{Man - hours worked}}$$

Note 1: If injury does not cause loss of time in the period in which it occurs but in a subsequent period, the injury should be included in the frequency rate of the period in which the loss of time begins.

Note 2: An injury causing intermittent loss of time, should be included in the frequency rate once at the first loss of time.

Note 3 – Frequency rate  $FR_A$  is used for all the comparison purposes because it uses actual number of lost time injury for the calculation. Frequency rate  $FR_B$  is used for all official purpose because it is based on the lost time injuries reportable to the statutory authorities.

### Severity Rate:

The method for calculation of severity rate is as follows:

$$SR_A = \frac{\text{Man days lost due to lost time injury} \times 10,00,000}{\text{Man - hours worked}}$$

$$SR_B = \frac{\text{Man days lost due to reportable lost time injury} \times 10,00,000}{\text{Man - hours worked}}$$

Severity rate  $SR_A$  is used for all the comparison purposes because it uses actual man days lost due to lost time injuries for the calculation. Severity rate  $SR_B$  is used for all official purpose because it is based on the lost time injuries reportable to the statutory authorities.

### Incidence Rates:

General incidence rate is the ratio of the number of injuries divided by number of persons for the period under evaluation. It is expressed as the number of injuries per 1000 persons employed.

The method for calculation of incidence rates is as follows:

$$IR_A = \frac{\text{Number of lost time injuries} \times 1000}{\text{Average number of persons employed}}$$

$$IR_B = \frac{\text{Number of reportable lost time injuries} \times 1000}{\text{Average number of persons employed}}$$

Incidence rate  $IR_A$  is used for all the comparison purposes because it uses actual number of lost time injury for the calculation. Incidence rate  $IR_B$  is used for all official purpose because it is based on the lost time injuries reportable to the statutory authorities.

## 13.7 ILLUSTRATIVE PROBLEM

### Example 13.1

A company has 200 full-time employees and 50 part-time employees and each works for 8 hours per day. Calculate Severity rate for the month of July, if number of injuries reported are 12 and if employees work for 6 days a week. Lost time for injury average is 20 days.

### Solution:

As number of employees are 250 working for 8 hours a day for a 27 days of a month, this equates to 54,000 manhours for month of July.

If the company experienced lost time injury of 20 days, then the formula works like this

$$\begin{aligned} \text{Severity Rate} &= \frac{\text{Man days lost due to reportable lost time injury} \times 10,00,000}{\text{Man hours worked}} \\ \text{Severity Rate} &= \frac{20 \times 10,00,000}{250 \times 8 \times 27} \\ \text{Severity Rate} &= \frac{20 \times 10,00,000}{54000} \\ \text{Severity Rate} &= 370 \end{aligned}$$

### SAQ 3

- What is frequency rate, severity rate and incident rate?
- Write the method for calculation of frequency rate, severity rate and incident rate?

## 13.8 INDUSTRIAL ACCIDENT AND INDIAN SCENARIO

Though, the industrial development has seen significant improvement in India, the situation of workers has not improved too much. Health and safety record of Indian industries is very poor. Since the beginning of liberalization, majority of regulations brought about were focused on facilitating the economic growth and development but on the contrary, condition of workers in industry has not improved much from the health and safety point of view.

It has been observed that, there are legislation to protect health and rights of workers' but their implementation is poor and thus only higher class of people in the organization ultimately enjoys the benefits. In India, only 8.8% workforce of the total available workforce is organized. The workforce in our country is abundant and easily available. Also, the majority of people in India

are unskilled and in addition, the high rate of unemployment makes them susceptible to exploitation. In our country it is difficult to get the work, thus workers think of getting an employment overlooking the hazard involved. It is found that unsafe working condition is one of the major cause of loss of human life and disability among India's workforce.

Report of the British safety council quoting the International Labor Organization numbers highlights that nearly 48,000 workers in India succumb to occupational injuries. Out of the total, a majority (24.20%) of victims are from the construction sector. The severity of this condition can be understood from the fact that in the year 2016 only 137 fatal incidents occurred in all the sectors of Britain; however, Figure for the fatal accidents in the construction sector in India is around 38 per day. Unfortunately, till date only 20 % workforce, out of 465 million, has been covered under the existing health and safety legal framework.

Some recent data revealed that 377 workers killed in accidents involved in mining of coal, minerals and oil between 2015 and 2017. Of the 377 deaths, 129 occurred in 2017 alone. As many as 145 died in 2016, while the Figure was 103 in 2015. Coal mines have accounted for the highest number of casualties due to accidents in mines. Of the 377, more than half, 210, were killed in coal mines.

**Some of the worst industrial disasters in India are listed below:**

**a) Maharashtra: Bombay docks explosion, 1944**

On 14<sup>th</sup> April 1944, the smoke was seen coming out of the vessel S.S. Fort Stikine in the afternoon, which was carrying the mixed cargo of ammunition for warfare and other flammable items like cotton bales, oil, timber and gold. Attempts were made to extinguish the fire, but it could not be controlled and spread swiftly due to the presence of highly flammable items in the vessel and reached to some ammunition causing two violent explosions that shook the Victoria Dock of Mumbai (then Bombay). The freighter carrying a mixed payload of warfare items and flammable got destroyed due to two heavy blast and fire. The impact of the explosion and fire was so severe that the other ships present in the area also sank after the explosion. It is believed that, it led to deaths of more than 800 people.

The magnitude of the two explosions was so high that it was recorded by the seismographs in the city at the Colaba Observatory. It took almost three days to control the fire. Later an enquiry was made on the incident which investigated several errors. One major reason was making the combination of commercial and wartime cargo in the same ship. Approximately 1400 tons of explosives were kept along with some flammable items like cotton, Sulphur and timber. Another reason identified was the delay in unloading the cargo. Ship arrived at Bombay port on 12<sup>th</sup> April 1944, but was still awaiting the

unloading at the time of blast. Timely unloading of explosives and flammable items could have avoided the blast and saved many lives.

**b) Jharkhand: Chasnala mining disaster, Dhanbad 1975**

In one of the India's as well as world's worst mining disasters, 372 miners died due to combined effect of explosion and resulting flooding from the nearby reservoir at the Chasnala coal mine near Dhanbad. The death claimed by the local worker's union is almost 700.

The explosion was most likely caused by sparks from equipment igniting a pocket of methane gas which is highly flammable. Even a small spark can ignite the surges of gas that may suddenly fill a mine. Clouds of coal dust raised by the explosion and accompanying shock wave contribute to these sorts of mine explosions, making the flames self-sustaining. Severity of explosion led to the total collapse of the mine and water from nearby reservoir rushed in to the pits at a rate of seven million gallons per minute.

**c) Chhattisgarh: Korba chimney collapse, 2009**

The incident happened on 23<sup>rd</sup> September 2009, when forty-five people, who were taking shelter from a thunderstorm, were killed due to collapse of an under construction chimney on them at the Bharat Aluminium Company (Balco) in Korba, Chhattisgarh. The plan was to construct a 275-meter-high chimney for the disposal of gases from the power plant. The construction was underway and reached the height of 240 meters. At the time of incident more than 100 workers were taking shelter in the nearby storeroom due to bad weather (heavy rains and lightning) when the under construction structure collapsed on top of the storeroom. More than 45 death casualties have been recorded in this incident.

**d) Rajasthan: Jaipur oil depot fire, 2009**

A major fire broke at Jaipur oil depot of Indian Oil Corporation (IOC) on 29<sup>th</sup> October 2009, leading to death of more than 12 people and wounding at least 200.

The fire started in an oil depot's giant tank holding 280,000 cubic feet of oil. The blaze was so intense and uncontrolled that it continued for more than a week. About 5 lacs people from the nearby area were evacuated after the incident due to intense fire and smoke. Petrol was being transferred from the IOC's depot to a pipeline at the time of accident.

**e) Uttar Pradesh: NTPC power plant explosion, 2017**

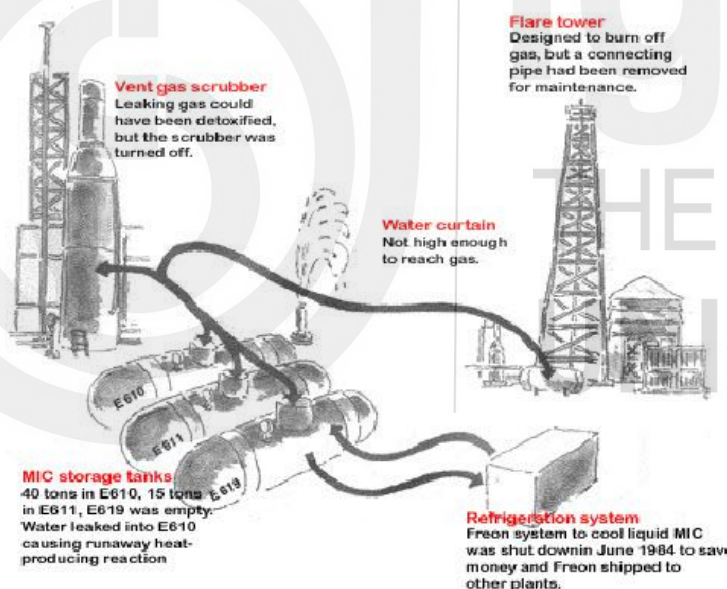
On November 1, 2017 a massive explosion occurred in a boiler at NTPC's Unchahar thermal power station plant in Uttar Pradesh's Raebareli district, leading to death of 34 people.

The major cause of the accident was identified as explosion inside the boiler. The common practice in the boiler operation is to regularly clear the ash deposited after the burning of coal. However, the head of operations, ash handling maintenance and boiler maintenance all went ahead with a decision to not to shut down the boiler for clearing the deposited ash which lead to the death of 34 people engaged in cleaning of burnt coal.

#### f) Bhopal: Union Carbide gas tragedy, 1984

This tragedy happened on the night of 2-3 December 1984, when around 40 tonne of methyl isocyanate gas leaked from a pesticide plant Union Carbide India Limited, Bhopal in Madhya Pradesh state. It is believed that, bout 30 metric tons of methyl isocyanate escaped from the tank into the atmosphere within an hour. The gas was blown by the wind into the surrounding slums and thousands immediately died of poisoning.

Even after so many decades of this massive gas tragedy, the picture on the number of dead and those affected is still unclear. The government list the death count as 5,295 however, activists Figures the death toll between 20,000 and 25,000.



**Figure 13.2: Events that Led to the Bhopal Disaster**

(Source: <http://www.hrdp-idrm.in/e5783/e17327/e24075/e27316/>)

The cause of the disaster remains under debate. The Indian government and local activists blame that the poor management and delayed maintenance was the basic reason for the tragedy. Deferring the maintenance of pipes caused the backflow of water into the MIC tank triggering the accident. It has also been found that most of the safety systems were not functioning and many valves were also in poor condition in November 1984. One of the major issue leading to the tragedy was that, the tank contained 42 tons of MIC, much higher than the safety limits of the tank.

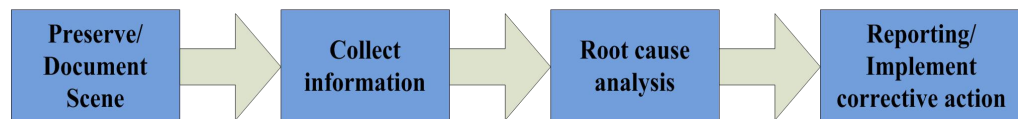
Bhopal gas tragedy is one of the worst industrial disaster of the world. It led to death of thousands of people overnight. Those who suffered the tragedy and survived are also facing the long term health effects.

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## **13.9 BASIC STEPS FOLLOWED IN ACCIDENT INVESTIGATION**

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An accident investigation involves three primary tasks, *i.e.*, gathering of useful information, analysis of the facts or possible reasons for the accident, and accident report writing. The steps involved in accident investigation are shown in Figure 13.3.



**Figure 13.3: Steps in Accident Investigation**

The investigation report is always intended to be a general guide for employers, supervisors, members of health and safety committee, members of an incident investigation team, etc. While investigating any accident, the main emphasis should be made on finding the root cause of the accident. Root cause analysis helps to avoid the occurrence of the same event in future by finding the facts that led to the accident, and measures that can lead to corrective actions. There are many reasons for investigation of any workplace related incident and it applies to all sort of incidents whether it is minor or major. Some of the major reasons are listed below:

- legal requirements of the country or state
- to find the root cause, so that such incidents can be prevented in future
- to determine whether the compliance with the safety rules were made or not
- to determine the cost of an incident on the company, so that insurance claims can be made
- to process the compensation/claims, if any, arising due to casualties

Thus, conducting an incident investigation is essential for any organization and should be conducted by the people who have thorough knowledge of the fundamentals of occupational health and safety rules and regulations and should have hands-on expertise of the following things:

- different causation models
- techniques of investigation
- knowledge of local and state government rules and regulations
- knowledge of the area of investigation like the design, operations, work flow, process flow, person and environmental impact
- knowledge of managerial techniques for questionnaire/ interview

- knowledge of different forms, documents, records, and data collection techniques
- knowledge of scientific methods for calculation to analyze the data for reporting and recommendations

The investigating officer forms a team and can include different people from inside/outside the organization to conduct the thorough investigation. Some of the members of the team generally includes; workers involved in the area, safety officer and some members from the safety department, employees having experience in incident investigation, representative of workers' union, experts from higher educations and officer from government and local police. The steps followed in accident investigation are as follows:

### **Step1: Preserve/document the Scene**

Preserving and documenting the scene is the first step towards incident investigation. The investigator should note each and every small thing related to the incident. Investigator should make sure that the scene has been captured through video recording, photographs and sketches so that anything left out can be revisited at the later stage. Also, the investigator should note all the losses incurred by the organization in terms of injuries to the employees, and loss of the material/machinery. All these things should be documented and preserved so that proper analysis can be performed at the later stage.

### **Step 2: Collect information**

The next step in incident investigation is collection of information related to the incident. It includes each and every small information related to the nature of the job, person involved in the operation, schedule of maintenance, last maintenance date, manuals of operations and maintenance, safety manuals, training reports, legal permissions, etc.

Thereafter, interview should be held for the various people in the organization related to that work as early as possible. An early interview of the people present at the time of incident can help investigator in collecting some very much useful and accurate information related to the incident because delaying interview may lead to influence of others as well as loss of information due to fade of memory.

The investigator should also make sure that the people injured or related to work do not get feared due to interview. Investigator should assure them that the purpose of the investigation is to find the facts and reasons of the incidence and not to blame anybody. It should be made clear that the purpose is to come to a solution to the root cause of the problem, so that such incident does not repeat in future.

### Step 3: Root cause analysis

A permanent solution to the problem cannot be reached until the root cause of the problem is determined. Root cause analysis is the deeper evaluation of the incident which helps investigator to reach the root of the problem associated with the incident. An incident can be due to the failure of the management, safety department, workers or the machine itself. Thus reaching to the exact reason to the problem ensures non-repetition of the same incident due to the same reason.

### Step 4: Reporting/ implementing corrective actions

Final steps to any investigation report is to propose some corrective actions based on the information gathered and root cause analysis. Report should be made very systematically and should clearly specify the root cause of the incident. Recommendations should be very specific, constructive, clear, easy to understand and feasible.

The reader of the report may have limited knowledge of the incident, so it is advisable to write a report in simple words, include all the events associated with the incident in the step by step manner, include photos and diagrams and clearly point the root cause of the incident. Also, mention the precautions or measures to be taken in future to avoid such incidents in the recommendation section.

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## 13.10 ELEMENTS OF INCIDENT INVESTIGATION FORMS

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Incident investigation form should be made in order to provide the complete information regarding and should consist of the following essential elements:

- **Specific date and time:** The form should have the correct and exact date and time of the incident.
- **Place and area:** Location of the incident should be mentioned in the form. Also, it should be very precisely mention the exact location or spot within the organization where the incident has occurred. Some maps and sketches should also be attached to the form for referencing and pointing the exact location. Pictorial representations are always useful in any report.
- **Employees injured:** The complete information regarding the number of employees injured, there name, department, job allotted, qualifications and contact details along with the medical history should be mentioned. Type of injury and body part affected due to the incident should be mentioned for each and every employee injured.
- **Damage to the property:** Accident investigation form should also contain the details of all the losses incurred to the organization due to the

incident. List of items should include all the machineries, electrical appliances, furniture, property damaged due to the incident.

- **Description of operation and incident:** This section should have details of the process, job activity of the employee, operating procedure, machine tools used, etc. After this, there should be thorough description of the sequence of events in step by step manner which led to the incident. Description by the employees involved in the job, where incident has occurred is very essential for preparation of this section. Employees' description of the incident should be properly documented.
- **Photographs taken and video recordings:** Photographs, video recording and sketches of the scene should be provided in the report.
- **Interview of the employees injured, involved in operations and witnesses:** Interview of the employees who have been injured or directly involved in the operation where incident has occurred provides complete in-depth picture of the incident. Witnesses, if any should also be interviewed to get any important information left out.
- **Supervisor/management acknowledgement:** Confirm that direct supervisors have acknowledged that an incident occurred by having them complete an incident investigation form promptly when the incident occurs. Fulfill management acknowledgement by having the person in charge of safety investigate the incident and ensure that the investigative report is processed/routed to the appropriate management chain.
- **Root cause analysis:** Root cause analysis report provides the actual cause of the incident and the most important part of any investigation form.
- **Preventive steps or any other recommendation for future:** Prepare a list of all the preventive steps required to avoid such incident in future. Safety recommendations and any mandatory change in the workplace or machinery if required should be highlighted. Expected date of completion of the mandatory safety rules, training of employees and implementation of recommendations suggested should be clearly specified.

#### SAQ 4

- a) Write a short note on some major industrial disasters in India.
- b) Write a short note on Bhopal Gas tragedy.
- c) What are the basic steps in accident investigation?
- d) Write different elements of incident investigation forms.

## 13.10 MODELS OF ACCIDENT CAUSATION

The models for studying causes and effects of accidents were initially developed to help experts examining them. The motive of these models is to research the causes of mishaps adequately. Prior knowledge on how the accidents are caused is valuable. It can help an investigator to proactively determine what sorts of failures or errors can cause accidents; thus, remedial course can be followed to address these failures before they get the opportunity to happen. In this regard, an Incident Ratio Pyramid (IRP) as shown in Figure 13.4 is created by experts in light of information from a wide range of industrial accidents. It is observed from IRP that every serious major injury is associated with a large number of minor injuries, events of property damage and incidents with no visible injury or damage. These incidents could be believed to show a fixed relationship. This idea has formed the basis of safety management systems. However, recently it has been brought to notice that different group of people have different views about the accident causation. Based on that some of the accident causation models have been developed. Note that process safety incidents are commonly less incessant, have more potential for injury, and 'near misses' are not as obvious. The barriers that should be vanquished to result in a process safety incident are likewise not quite the same as those which are pertinent for an occupational safety incident.

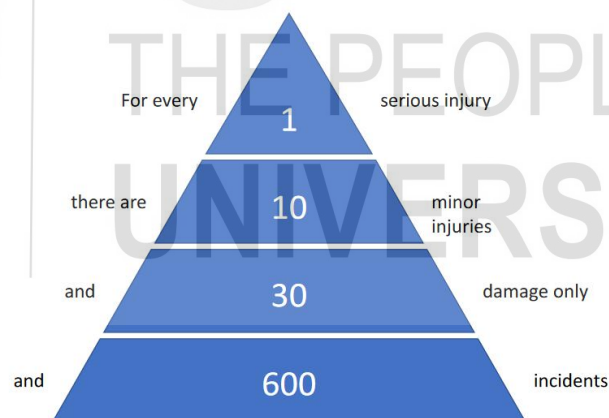


Figure 13.4: Incident Pyramid

Accident models provide a conceptualization of the characteristics of the accident, which typically show the relation between causes and effects. They explain why accidents occur, and are used as techniques for risk assessment during system development, and for post-accident analysis to study the causes of the occurrence of an accident and further measures to control the accidents. Most of the engineering models originated before the introduction of digital technology; these models have been updated but have not kept pace with the fast change in technological revolution. Modern technology is having a significant impact on the nature of accidents, and this requires new causal explanatory mechanisms to understand them and in the development

of new risk assessment techniques to prevent their occurrence. Different theory involved in models of accident causation are as follows:

## 1. The Domino Theory

Herbert W. Heinrich was a pioneering occupational safety researcher, most famous for originating the concept of the “safety pyramid”. Heinrich’s domino theory postulates that accidents result from a chain of sequential events, metaphorically like a line of dominos falling over. When one of the dominos falls, it triggers the next one, and the next and so on. According to Heinrich, 88% of all accidents are caused by unsafe acts of people, 10% by unsafe actions and 2% by “acts of God”. He proposed a “five-factor accident sequence” in which each factor would actuate the next step in the manner of toppling dominoes lined up in a row as shown in Figure 13.5.

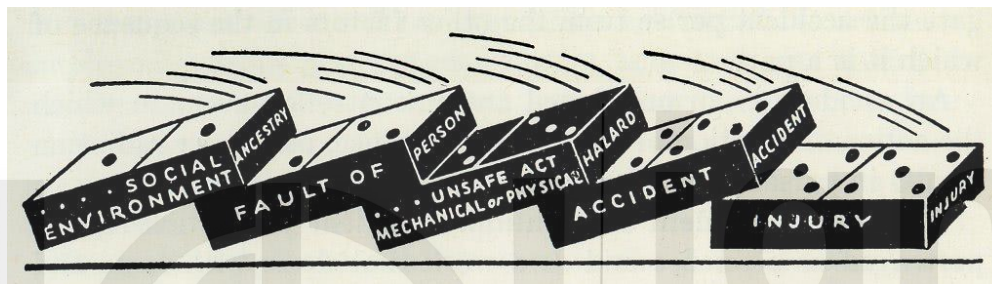


Figure 13.5: The injury is caused by the action of preceding factors

(Source: Heinrich HW. Industrial Accident Prevention. A Scientific Approach. (Second Edition))

Heinrich posits five metaphorical dominoes labelled with accident causes. They are Social Environment and Ancestry, Fault of Person, Unsafe Act or Mechanical or Physical Hazard (unsafe condition), Accident, and Injury. Heinrich defines each of these “dominoes” explicitly, and gives advice on minimizing or eliminating their presence in the sequence.

**domino 1 (Social Environment and Ancestry):** ancestry and the worker’s social environment, which impact the worker’s skills, beliefs and “traits of character”, and thus the way in which they perform tasks

**domino 2 (Fault of Person):** the worker’s carelessness or personal faults, which lead them to pay insufficient attention to the task. Inborn or obtained character flaws such as bad temper, inconsiderateness, ignorance, and recklessness contribute to accident causation.

**domino 3 (Unsafe Act and/or Unsafe Condition):** an unsafe act or a mechanical/physical hazard, such as a worker error (standing under suspended loads, starting machinery without warning...) or a technical equipment failure or insufficiently protected machinery

**domino 4 (Accident):** events such as fall of persons, striking of persons by flying objects are typical accidents that cause injury.

**domino 5 (injury):** injuries or loss, the consequences of the accident. Some types of injuries, Heinrich specifies in his “Explanation of Factors”, are cuts and broken bones.

## **2. The Pure Chance Theory**

As per the pure chance theory, for a given random set of workers each one of them has an equivalent possibility of being engaged in a mishap. It further infers that there is no particular perceptible pattern of events that prompts a mishap. In this hypothesis, all mishaps are treated as corresponding to Heinrich's acts of God, and it is held that there exist no intercessions to avoid them.

## **3. The Biased Liability Theory**

Biased liability theory is based on the view that once a worker is engaged with a mishap the odds of him/her getting to be associated with future mishaps are either expanded or diminished in comparison to the remaining workers.

The contribution of biased liability theory in developing preventive actions for avoiding accidents is minimal!

## **4. The Accident Proneness Theory**

Accident proneness theory is based on the view that within a given set of workers, there exists a section of them who are increasingly at risk of being engaged with mishaps. The credibility of this theory is questionable as researchers have not been able to demonstrate its correctness conclusively due to poorly conducted experiments, and inadequate, uncertain and conflicting outcomes. This theory is not largely recognized. It is felt that if indeed this theory is supported by any empirical evidence at all, it probably accounts for only a very low proportion of accidents without any statistical significance.

## **5. The Energy Transfer Theory**

Energy transfer theory is based on the claim that, the injury of the worker or damage to the property of the organization is the result of change of theory. The concept of energy transfer theory states that for every change of theory there is source, a path and a receiver. This theory is useful for determining injury causation and evaluating energy hazards and control methodology. It helps in development of strategies which either prevents, reduces or vanishes with respect to the energy transfer.

**Source:** It is the first point where control of energy transfer can be achieved. It can be achieved by:

- removal of the source itself

- making the changes made in design/ specification of elements of the work station
- proactive maintenance.

**Path:** It is the second point where control of energy transfer can be achieved. It can be achieved by modifying the path of the energy as follows:

- complete enclosure of the path
- abrupton of the path by installing various barriers
- installation of some absorbers which can absorb the energy
- Placing of isolators.

**Receiver:** This is the end point of the system where control of energy transfer can be achieved. It can be achieved by adopting the following measures:

- limitation of exposure
- use of safety gears for protection of individual

## 6. The Multiple Causation Theory

This theory/model is very similar to the domino theory. It claims that there are many causal elements behind each accident. In other words, any accident occurs as a result of combinations of various primary causal elements and sub-elements. As per this model, the causal elements can be categorized as:

- 1) Behavioural: Comprising of elements like negligent behavior, lack of knowledge, limited skills and poor physical and mental condition of the work-force.
- 2) Environmental: Including insufficient mechanisms for protection against various hazardous work elements, and wear-out of instruments/machines with regular usage and unsafe work practices.

Thus, this theory confirms that very rarely an accident happens because of a single causal element.

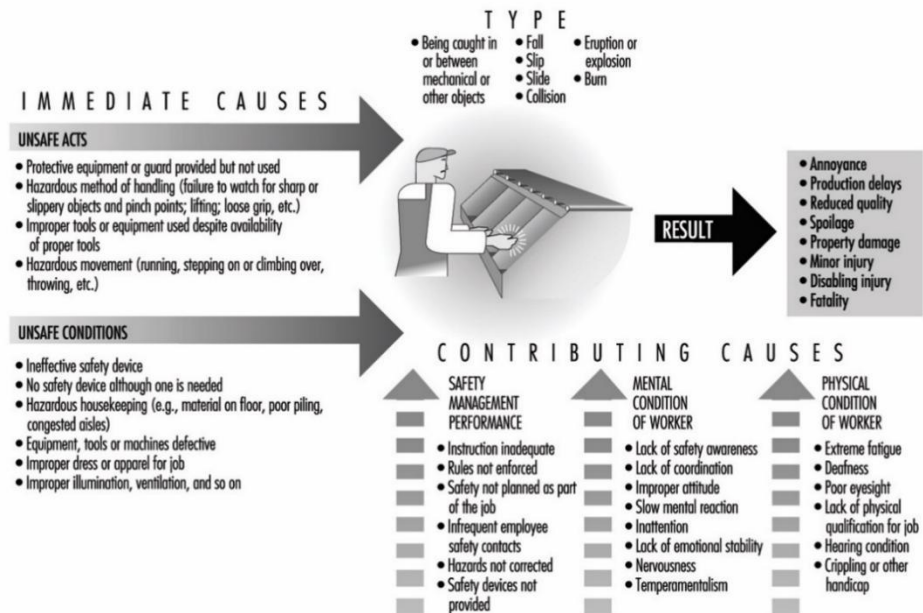
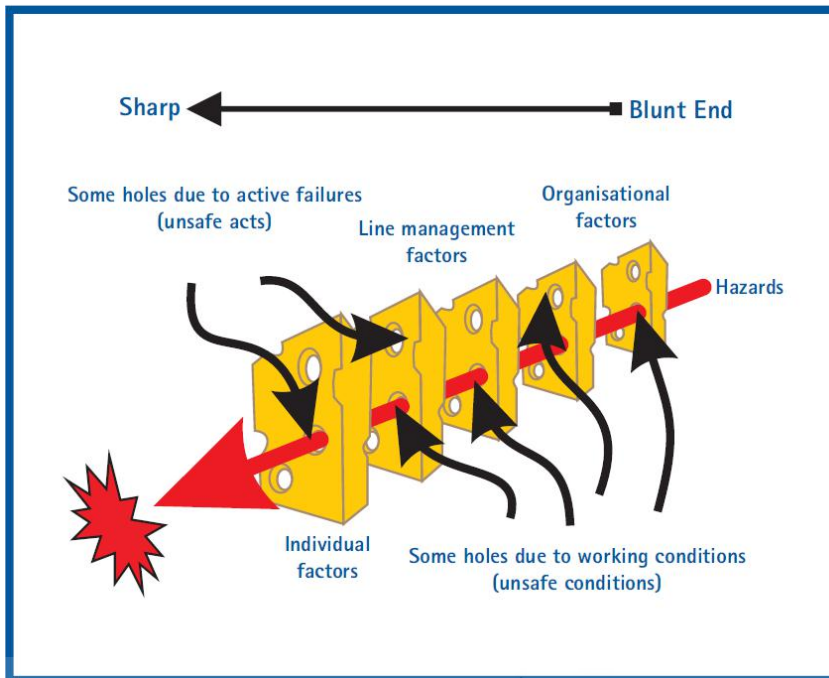


Figure 13.6: Types and Causes of Accident

(Source: <http://www.ilocis.org/documents/chpt56e.htm>)

## 7. Swiss cheese theory

According to this model, albeit existence of numerous protection layers among risks and accidents, there may be concealed weaknesses. These weaknesses, in each layer, if aligned can trigger an accident. For instance, refer Figure 13.8 wherein all, side by side, stacked slices represent the protection layers and the holes within, depict concealed weaknesses. The holes may be present due to the deficiency in different parts of the system. Also, these deficiencies or flaws may be of variable size, and may be present at different locations of each slice. Thus, above defined, each stacked layer acts as a barrier to avoid the danger of a potential mishap turning into a reality. However, even in the presence of these barriers, still there are chances of occurrence of an accident due to the alignment of holes. The momentary alignment of holes presents an opportunity to hazards to pass through all stacked layers, and thus cause a failure or an accident! The studies related to this model suggest that most errors are caused by an organization's ignorance for having complete and robust layers of protection; this enables weaknesses to assemble and "cross the cheese" to cause an accident.

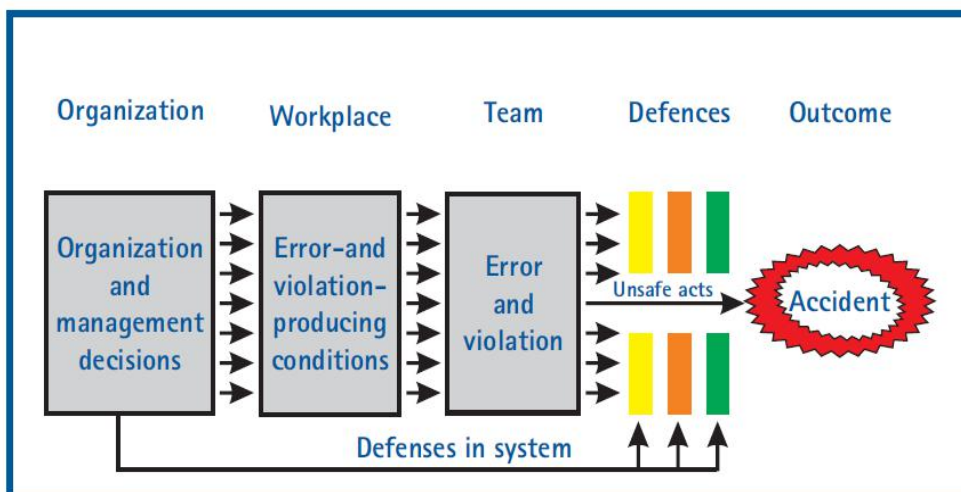


**Figure 13.7: Reason's Swiss Cheese Model**

(Source: <http://www.hrdp-idrm.in/e5783/e17327/e24075/>)

## 8. Modern theory of accident

As per modern accident theory, accidents are cumulative effect of aggregation of various factors which have potential to cause mishap(s), but in itself are inadequate to do any large scale damage. In many instances, the equipment failure or operator errors are rarely the only reason behind breach in safety mechanisms. Usually, such breakdowns are the outcome of failure of the involved human being in decision-making. The breakdowns may include active failures at the operational level, or dormant conditions having a potential to cause a breach in the system's safety mechanisms. Often, mishaps comprise of both active and dormant flaws. Refer Figure 13.8, which depicts a model to help us understand the relationship between organizational and management factors (*i.e.*, system factors) in accident causation.



**Figure 13.8: Modern Theory of Accident**

Various “defences” are incorporated into the plant system to protect against inappropriate performance or poor decisions at all levels of the system. It includes the workers directly involved in the job, the supervisor for the department, head of the department and top management. This model of modern theory, shows that all these factors can create latent conditions leading to an accident and also contribute to the system's defences.

Any violation of the recommended procedure or error in work are the unsafe acts and have the immediate effect on the system, and may penetrate through various defences which are used as a protection gears for the system to operate smoothly in any organization. Unsafe acts are generally the results of error, but sometimes the result of overlooking the rules and intentionally breaking them in order to achieves some short term gains.

These unsafe acts are committed in the working conditions and thus have hidden but possible unsafe conditions. Such a latent unsafe condition is the result of wrong decisions and may not have immediate effect on anybody due to its inactive behaviors. But the consequences of this type of unsafe condition could harm at the mass level once the defence system is breached.

It has been observed that these unsafe conditions are the results of the wrong decisions of the management. Senior management works under various restrictions related to time, money, material, government policies, demand from the customer, etc. Combined effect all these things make the decision maker sometime biased and ending up with the latent unsafe conditions.

Thus, effective maintenance and safety measurements are vital for finding and alleviating the laten unsafe conditions from the system to minimize the chances of any incident. Effective safety management efforts aim to identify and mitigate these latent unsafe conditions on a system-wide basis, rather than by localized efforts to minimize unsafe acts by individuals. Such unsafe acts may only be symptoms of safety problems, not causes.

#### **SAQ 5**

- a) What is accident causation model and why they are developed?
- b) Draw and explain incident pyramid.
- c) Write the name of any six accident causation model.
- d) Explain in brief the Domino theory and the Swiss Cheese theory of accident causation.

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## 13.12 LET US SUM UP

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Accident is an unexpected event in the course of employment which is neither anticipated nor designed to occur. Thus, an accident is an unplanned and uncontrolled event in which an action or reaction of an object, a substance, a person, or a radiation results in personal injury. It is important to note that self inflicted injuries cannot be regarded as accidents.

An industrial injury is defined as “a personal injury to an employee which has been caused by an accident or an occupational disease and which arises out of or in the course of employment and which could entitle such employee to compensation under Workers’ Compensation Act, 1923”.

Accidents may be of different types depending upon the severity, durability and degree of the injury. An accident causing death or permanent or prolonged disability to the injured employee is called ‘major accident. A cut that does not render the employee disabled is termed as ‘minor’ accident. When an employee gets injury with external signs of it, it is external injury. Injury without showing external signs such as a fractured bone is called an internal one.

When an injury renders an injured employee disabled for a short period, say, a day or a week, it is a temporary accident. On the contrary, making injured employee disabled for ever is called permanent accident. Disability caused by accident may be partial or total, fatal or non-fatal.

An accident investigation involves three primary tasks, i.e., gathering of useful information, analysis of the facts or possible reasons for the accident, and accident report writing. It is observed from IRP that every serious major injury is associated with a large number of minor injuries, events of property damage and incidents with no visible injury or damage. These incidents could be believed to show a fixed relationship. This idea has formed the basis of safety management systems. However, recently it has been brought to notice that different group of people have different views about the accident causation. Based on that some of the accident causation models have been developed.

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## 13.13 KEY WORDS

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**ACCIDENT:** An incident which results in death, injury loss, or damage.

**ACCIDENT TRIANGLE:** Indicates statistical relationship and severity of accident.

**DANGER :** A state or condition in which personal injury and/or asset damage is reasonably foreseeable. The presence of a hazard.

**NEGLIGENCE:** The omission to do something, which a reasonable person, guided upon those considerations which ordinarily regulate the conduct of

human affairs would do, or something, which a prudent and reasonable person would not do.

**SAFETY:** Freedom from (unacceptable) risk of harm to persons. Safety may also encompass environmental or asset damage/loss.

**SEVERITY:** of a hazard, the degree of harm which a hazard can create if it occurs; the measure of severity depends on the industry sector.

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### **13.14 ASWERS TO SAQs**

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For all SAQs, please refer to the relevant text in the unit.

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### **13.15 REFERENCES AND FURTHER STUDIES**

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## UNIT 14 TYPES OF ACCIDENTS AND ITS ANALYSIS

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### Structure

- 14.1 Introduction
  - Objectives
- 14.2 Key Factors of Accident Analysis
- 14.3 Purpose of Accident Analysis
- 14.4 Simple Techniques of Accident Analysis
- 14.5 Advanced Techniques
- 14.6 Types of Investigations and Analysis of Accident
- 14.7 Basic Components of Accident Chains for Analysis of Accident
- 14.8 Case History: Jaipur oil depot fire-2009
- 14.9 Let Us Sum Up
- 14.10 Key Words
- 14.11 Answers to SAQs
- 14.12 References and Further Readings

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### 14.1 INTRODUCTION

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Accident is an undesired event. Accidents results in a physical injury or property damage or both, usually due to contact with a source of energy which is above the ability of the body or structure to withstand it. According to Ridley (1986), 99% of the accidents happen are due to either unsafe acts or unsafe conditions or both. Accidents are avoidable and should always be avoided irrespective of the level of damage or los they can cause. Each and every individual as well as industry as a whole should take all necessary steps to avoid and minimize the accidents in future. Recent studies suggest that around 2.3 million laborer mortalities occur due to work-related accidents and diseases. More than 474 million people endure chronic ailments, resulting from occupational activity, and non-fatal accidents. The fatality rate from construction accidents are among the highest compared to the overall industry. Studies suggest that hazardous working conditions contribute a meager 3 % among all other type of workplace accidents. Negligent attitude (unsafe act) of the worker account for 95 % of all workplace accidents. Uncontrollable acts add 2 % of all workplace accidents

Safety is a responsibility, requirement and asset of all personnel in the industry. The accident caused is the action or the condition immediately preceding the accident. From this we can observe that accidents are caused by unsafe conditions and by unsafe acts.

Hence it should be a fixed rule that accidents be reported at once and investigated immediately. Near-accidents or accidents not resulting in injury also should be investigated. The purpose of accident investigation is to get at those factors which if eliminated will prevent a reoccurrence.

Accident analysis is an analysis which is carried out in order to determine the cause(s) of an accident so as to make best effort to prevent further incidents of a similar nature in future. It is also known as accident investigation or incident investigation.

### **Objectives**

After studying this unit, you should be able to

- identify the major elements of good investigation & analysis.
- classify the types of investigations for the accident analysis
- discuss the essential theories and tools for the analysis of accident.
- describe the components of accident chains for analysis of accident.

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## **14.2 KEY FACTS CONSIDERED IN THE ANALYSIS OF AN ACCIDENT**

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Investigation of the accidents is primarily regarded as a fact-finding procedure; which are based on the facts that were not followed which could also be prevented in order to avoid recurrences of similar accidents. If properly handled, accidents can also be reduced and that enables the safety and health awareness among the employees, and builds affinity between the supervisor and the injured. These injuries and also the illness among the workers can be reduced to great extent by minimizing the hazardous conditions. Although it is to be remembered that it cannot be completely eliminate all accidents. However, it will be easier and cost effective, if it can be reduce and eliminate the hazards than to teach employees to be safe from being injured in the working area. It is significant to note that reducing or minimizing the hazards is important. This can be better accomplished by changing the worn parts and equipments, installing the safety guards, by enhancing the material handling technique, and by adopting less hazardous chemicals.

It is also important to rectify the causes and adopting the corrective measures as is the objective of any accident investigation procedure. Examples of such areas to be probes include unsafe conditions, poorly underlined tasks, inadequate procedures and lack of accountability. Even the minor injuries that occurred on the floor evoke the usage of incorrect practices, which if corrected could be used to avoid such hazards. Therefore, every minor injury should also be considered and investigated in the same manner as vigor with thoroughness.

Followings are the key facts of any accident analysis:

**a) Nature of Injury**

It is the type of a physical injury. If two or more types of injuries are sustained, one which is more severe than the other should be selected for reporting and injury causing permanent disablement may be selected in preference to temporary injury.

If the injury is confirmed to the part of the body that part should be mentioned. But if it extends over major body part that major body part to be mentioned. For internal injury the part affected by the accident should be mentioned.

**b) Source of Injury**

The object, substance or bodily motion directly responsible for the injury should be mentioned in the report. If the injury result from the contact with two or more objects, say a moving object and a stationery one, the moving object should be considered in a report.

**c) Accident Type**

The accident type depends on the source of injury classification. The contact which affects the victim most should be taken as the type of accident. Generally the accidents are classified into three categories on the basis of the length of the recovery:

**1. First Aid Cases:**

The injuries happened due to minor accidents are not considered to be serious as the worker is given the first aid in factory hospital. As the worker gets the medical treatment in the factory hospital can return to start the work. This will not be time consuming except for the time taken to treat the worker with the first aid. Also, there will be no compensation being paid due to minor injuries.

**2. Home case accidents:**

Preliminary aid in treatment is being at the factory hospital and allows the worker to go home for the fast recovery. During this wound will be healed and the worker will be ready to resume his duties. Therefore, worker may lose a shift or day of his duties during the accident. This type of accidents also not serves the purpose of compensation for the injured as this will not fall under workmen's compensation act.

**3. Lost Time Accidents:**

These accidents are entitled for the compensation from the company. The worker will be allowed to take leave on the work on the issue of accident for a day and entitled to take many days if necessary. During this the worker will

be hospitalized as a result of temporary or a permanent disablement might occur.

This leads to an enquiry followed by the investigation if there exists a difference in causes is found concerning the accident. As an example, if the body part such as hand, leg or any other part is injured or being cut.

#### **d) Hazardous Condition**

The condition or circumstance what directly caused the accident is to be named in this classification. The physical condition or circumstances that permitted or occasioned the occurrence of the accident. This generally takes into account the accident type and the agency of the accident.

Employees in an industrial establishment are often subject to certain health hazards and occupational diseases. According to Roland Black, the normal occupational health hazards may be classified into:

1. Chemical Hazard
2. Biological Hazard
3. Physical Hazard
4. Mechanical Hazard
5. Psychological Hazard

#### **e) Accident Cause:**

The accidents could occur as a result of faulty equipment or machinery and also due to the negligence of the worker. These accidents can be reduced with proper safety measures and precautions. There will always be some of the valid reasons for these accidents to happen. Also, chances of occurring accidents during running the machine and equipments are also quite often. Therefore, proper training with the prior knowledge among the workers can reduce the danger of accidents as operation in industry is always a full chance of accidents. Carelessness among the workers also leads to accidents. Broadly these causes are classified in two categories:

#### **Unsafe Acts:**

It is the human action that departs from a standard job procedure or safe practice, safety regulations or instructions. An unsafe act is failure to do something which a person should have done or to do something differently which he ought not to do. Generally the unsafe acts are:

- trying to operate an equipment without authority
- not adhering to standard operating procedures and instructions
- either not using the personal protective equipment or removing them intermittently
- using defective equipment or incorrect tool

- using improper PPEs or using PPEs improperly
- horseplay
- not taking care of ergonomics and working in an unsafe posture
- absent-mindedness / nervousness
- willful intent to injure
- using any type of intoxication at the work place
- improper lifting and carrying or improper tools
- lack of knowledge or skills
- not following the instructions or not able to understand the same

**Unsafe Conditions:** The unsafe condition is a hazardous physical condition or circumstances permitting the occurrence of an accident. It could be the physical or chemical property of a material, machine or the environment which could result in injury to a person, damage or destruction to property or other forms of losses. Some of them are given below:

- Wet floors or slippery floors
- Unstable stacking of materials
- Protruding re-bars
- Improper insulation on any live conductive material
- Equipment without machine guarding
- Poor storage of combustible materials

**f) Agency of an accident**

The object, substance or part of the premises in which the hazardous condition exist should be named. Condition exist should be named. It may not be identical with the source of injury. The source of the injury is one which directly causes the injury irrespective of whether it was hazardous at all.

If a particular part of agency is distinctly hazardous and not the rest of it, the part should be named.

**SAQ 1**

- a) What are the key facts considered in analysis of accident? Explain!
- b) Write short notes on:-
  - (i) Immediate causes of accident
  - (ii) Agency of accident
  - (iii) Hazardous condition

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## 14.3 PURPOSE OF INVESTIGATION AND ANALYSIS

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An incident investigation is a structured or systematic procedure that includes the recording and review in the form of documentation and analysis, including a detailed evaluation of contributing factors, of a workplace accident that resulted in a loss or possible loss. The purpose of the incident report is to produce suggestions that will remove the risk for possible damages. The purpose of performing incident investigations becomes necessitate due to:

- Fulfillment of legal requirements
- Assessing the compliance of the operating procedures with established safety procedures
- Preventing the occurrence of future accidents of similar nature
- Enabling the processing of any worker's compensation claims arising due to any injury
- Demonstrating concern of the organization for employee well-being

There is difference in an accident and incident. An accident can be described as an unintended event that results in injury or property damage. An incident can be described as an unintended event that does not result in injury or property damage. Additionally, a hazard can be described as the potential to do harm, while risk is defined as the likelihood of harm actually occurring. Other terms such as "Near miss" are used to describe incidents that could have easily ended up being serious accidents. Some jurisdictions avoid the use of the term "accident", because the latter term infers that the event could not have been avoided. They choose to use "incident" instead, pointing out that most events are predictable and preventable. To the extent that the risk is managed, there is less chance of an accident occurring. The depth of the investigation should be appropriate to the seriousness of the situation in terms of actual or potential injury.

The investigation process includes six important steps. These include:

- Investigation of the accident
- Identification of direct and root causes
- Disclosure of findings
- Development of a corrective action plan
- Implementation
- Follow-up review and revision as necessary

There are generally five major elements in a good accident investigation and analysis. The Six W's is one simple but proven methodology that helps field investigators build a solid understanding of the event including Specific

procedures, conditions, unsafe elements (acts and/or behavior) and corrective actions. These five elements are:

- How, when, where and why did the accident occur?
- What procedures were being followed and why?
- What conditions existed when the accident occurred?
- What was the primary / basic cause of the accident?
- What should be done to prevent similar occurrences?

All accidents are caused and could have been prevented by the identification and removal of one or more of the contributing factors. All possible factors must be discussed and identified. Participants in the investigation should agree with the conclusions. Each accident is to be treated as a matter of real importance. Reporting must be encouraged and investigation must be prompt. Supervisor and employee must work as a team to identify accident causes and corrective actions needed to prevent similar accidents. Each accident results from a breakdown in the safety system which includes any of the following:

- Employee behavior
- procedures
- the condition of equipment
- environment

The ability to monitor and evaluate these elements on a continuous basis can identify work practices and conditions that have the ability to produce accidents. Most injuries involve the workshop employees who “do the work. Whenever the potential to cause an accident / injury is apparent, corrective actions must be implemented.

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## **14.4 SIMPLE TECHNIQUES**

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Accident is always undesirable. It is generally conceived that accident investigation requires a person to be always a safety professional. However, there are some simple investigation techniques which do not necessitate a user to be a safety professional, i.e. learning these techniques do not really require a long period of training or a certified degree. It only requires commitment and orientation towards learning the simple but effective techniques. A typical feature of such a simple technique is that, such a technique will not take excessive time. On an average it will consume just couple of hours to perform accident investigation.

One of the well known techniques for accident investigation is considered to be a Finnish model. The Finnish model for accident investigation is not mandatory legal requirement, but it is a realistic and easy-to-use method for workplace accident investigation that non-experts can use and benefit.

In the Finnish model it is proposed that accident investigation should be carried out in working groups containing individuals from various levels of the organisation. Answers to following questions helps in accident investigation:

- What happened (description)?
- Where did it happen?
- What were the circumstances at the accident scene?
- Which persons, machines, equipment were involved in the accident?
- What work was being performed when the accident occurred?
- Was there anything unusual in the situation?

There are ten basic steps in Finnish model for investigating occupational accidents:

1. Orientating to the accident case: After the accident, to obtain information about what happened, it is extremely important to check the scene immediately and should be the very first and quick step. It is important to interview eyewitnesses and photograph the circumstances. It is necessary to identify and record all odd and deviant incidents and occurrences.

Following points should be included in inspecting the scene of the accident.

- i) Complete details of the victim(s) along with location, eyewitnesses, and other persons who were working in the area;
  - ii) What was being done and details of the equipment being used;
  - iii) The operating conditions as well as the circumstances at the accident scene;
  - iv) The operating conditions and circumstances of the wider working environment in general (i.e. lighting, noise, and etc.);
  - v) The skill set and level of training provided to the personnel involved;
  - vi) Organisation of the work being allocated and other responsibilities of the persons involved.
2. Describing all the events occurred during that day in chronological order: The events should be outlined and separated– An easy way to undertake this description is to start with the accident itself. The investigation should be extended backwards until the last "normal" working act was performed, thus it is not enough to describe only to the event that led to the accident:
    - What were the various previous events in chronological order before the actual accident occurred?

- What was the end result of the events? (injury type and injured body part)
  - What was the type of the accident?
  - What was the concrete cause of the injury?
3. Collecting concrete data on how the victim was involved in the cause of the injury:
    - The scene or place where accident has occurred and occasion;
    - What was he/she actually doing before the accident.
  4. Collecting details about how the cause of the injury was connected to the cause of the accident:
    - As part of normal operations, the cause of the accident may occur, but it may also be caused by damaged or malfunctioning machines and/or equipment, or equipment improperly installed in the workplace.
  5. Collecting information on all contributing factors, i.e. what were the various factors that contributed to the accident:
    - Contributing factors, as provided in step 2, should be considered for each event;
    - It is possible for each event to have more than one contributing factor;
    - Identifying contributing factors is based on close inspection, rather than just speculation behind the office desk, of the real injury method.
  6. Collecting information on why did the cause of the injury exist and how did it come to be present at the accident scene, especially when it is not its permanent location. One should also consider what were the accident factors contributing to the existence of the cause of the injury.
  7. Considering the possible ways to prevent occurrence of similar accidents.
  8. Choosing the appropriate steps for the future prevention of similar incidents and considering how best to enforce the following actions:
    - If there are many optional steps, it is necessary to consider which one is the best and most feasible one for implementation.
    - Select the person in charge of implementing these measures;
    - Set an implementation timetable.
  9. Providing and distributing details on the findings of a workplace accident investigation to all the employees:
    - In addition to those at the site of the accident, all authorities and different departments in the organization must also be notified, since similar incidents can also occur in other places.

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## 14.5 ADVANCED TECHNIQUES

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There are some advanced techniques for systematic investigation of accident. Good examples of some of the more complex but systematic accident investigation techniques are AcciMap, MTO-analyses model, STAMP model, and FRAM method. Until being learned in practice, each of these advanced techniques requires specialised training and will therefore only be briefly overviewed in this article.

### The Acci Map

The AcciMap methodology for accident analysis is based on the risk management method by Rasmussen. Different accident situations are initially chosen and a cause-consequence map is used to evaluate the causal chains of events. A cause-consequence map reflects a generalisation that incorporates a series of incidental event courses. As the basis for predictive risk analysis, outcome charts have been widely used.

The set of input that is chosen to be included in a cause-consequence chart is defined by the choice of the critical event, which reflects the release of a well-defined hazard source, such as “loss of containment of hazardous substance”, or “loss of control of accumulated energy”. The critical event in the problem links the causal tree (the logic relation among potential causes) with the subsequent event tree. The model thus serves to define specific decision-makers and the usual working situation in which future incidents are influenced and modulated.

AcciMap focuses not on the conventional quest for identifying the "guilty person," but on identifying those individuals in the system who can make decisions that lead to improved risk management and, thus, to the design of improved system protection. Thus, the intention is not to blame anybody but to identify the actual cause and the ways to mitigate the risk in future.

### STAMP

STAMP stands for Systems Theoretic Accident Modeling and Processes. In safety management, it focuses on the role of constraints. Safety is defined as a continuous control task instead of defining safety in terms of preventing component failure events to enforce the constraints required to restrict system behaviour to ensure only safe changes and adaptations. Accidents are seen to arise from insufficient supervision or implementation of safety-related conduct constraints at each level of system growth and control structures of system operations. Accidents can also be interpreted in terms of why the safeguards in place did not avoid or identify maladaptive changes ( e.g. the detection of the safety constraints that were broken at each stage of the control mechanism, as well as why the constraints were insufficient or why

the system was unable to exercise adequate control over their compliance if they were theoretically adequate).

The process of an accident which leads to loss event, can be defined as an adaptive feedback function that fails to maintain safety of the employee to meet a complex set of goals and values as output changes over time. This adaptive feedback mechanism enables the model to integrate adaptation as a fundamental property.

### **MTO-analysis**

MTO analysis stands for Man, Technology and Organization. The rationale for the MTO-analysis is that, it emphasizes that in an accident investigation, human, organisational and technological factors are equally important and thus should be taken care. The approach to carry out this is based on the "Human Performance Enhancement System (HPES).

There are three different methods for carrying out the MTO-analysis:

1. Structured analysis- this uses the event- and cause-diagram for investigation;
2. Change analysis - examination of transition by outlining how incidents have deviated from past events or standard practise;
3. Barrier analysis- finding failed or missing technical and administrative barriers.

The first step in an MTO-analysis is to establish a sequence of events in longitudinal manner and to demonstrate a block diagram of the event sequence. The next step is to classify and vertically draw potential technological and human triggers of each event into each event in the diagram. The third step is to examine the technological, human or organisational barriers that failed or were missing during the accident and to explain all the barriers below the incidents that were missing or failed.

The MTO-methodology also provides a checklist for the identification of failure causes. The following considerations are included in the checklist: job structure, work practice, work management, processes for transition, technology ergonomics / deficiencies, communication, instructions / procedures, education / competence, and work environment. There is a comprehensive checklist of essential or essential causes for each of these causes of failure.

### **Functional Resonance Accident Model (FRAM)**

The Functional Resonance Accident Model (FRAM) and the related approach offer a way to explain how to combine several functions and conditions to create an adverse effect.

FRAM is based on the following principles:

1. The Principle of equivalence of success and failure: FRAM adheres to the view of resilience engineering that failures reflect the reverse side of the modifications required to deal with the complexities of the real world rather than a failure of normal functions of the system. Success depends on the capacity of organisations, groups and individuals to predict threats and to consider, identify, and take effective action in critical situations; failure is due to the temporary or permanent absence of such ability.
2. The principle of approximate adjustments: Since the working conditions never fully conform to the conditions defined or recommended, individuals and organisations must always change their performance in order to function, specifically the actual resources and requirements, under the current conditions. Since resources ( time, manpower, data, etc.) are often limited, these modifications are invariably approximations rather than precise characteristics.
3. The principle of emergence: Normal performance variability is seldom large enough to be the cause of an accident in itself or even to constitute a malfunction. The variability of multiple functions, however, can combine in unpredictable ways, leading to unreasonably large consequences; thus, they generate a non-linear effect. Instead of resulting phenomena, both failures and normal performance are emerging, since neither can be attributed to or explained only by referring to the (mal) functions of particular components or parts.
4. The principle of functional resonance: Every now and then, the variability of a number of functions can resonate, i.e., strengthen each other and thus lead to variability so that one function exceeds the usual limits. As explained by the Small World Phenomenon, the consequences can spread via tight couplings rather than through observable and enumerable cause-effect connections. This can be defined as a resonance, hence functional resonance, of the normal variability of functions. The comparison of resonance emphasises that this is a complex phenomenon, and thus not due to a clear combination of causal connections.

The clarification is produced while conducting an accident investigation with FRAM by going through the following steps: Step 1. Identify vital system functions as a reference, using normal or accident-free performance. This phase separately characterises each feature, but does not attempt to organise or order them in any way. An current task analysis, processes, expert expertise, etc., can be the starting point. The following six elements are used in the characterisation:

- Input (I): the element which is processed or transformed by the function,
- Output (O): the element which is the result of the function based on the inputs, it is the change in entity or state,

- Preconditions (P): conditions that must exist before a function can be executed,
- Resources (R): that which the function needs or consumes to produce the output,
- Time (T): temporal constraints affecting the function (with regard to starting time, finishing time or duration),
- Control (C): how the function is monitored or controlled.

The following steps are followed:

**Step 1.** A simple table, which can then be used for further study, can define each function.

**Step 2.** Considering both current and possible uncertainty, describe the observed variability of system functions. The goal of FRAM is to provide an interpretation of the accident in terms of output variability combinations. Therefore, for each feature, the second step is to identify the actual variability during the accident. This could point to other functions which, as part of the clarification, must be characterised. For example, if the input to a function came too late or was of the wrong type, then it is important to define and characterise the source of that input, i.e., another function. In turn, this could involve defining still more roles before one has accounted for the complete scenario.

**Step 3.** Identify and explain the functional resonance between functions and observed output variability from the observed dependencies / couplings. A list of functions, each distinguished by two or more of the six aspects, is the product of the first and the second phases. (Note that a function can involve the description of many instances of an aspect.) By matching or linking their aspects, the dependencies between functions can be identified. For example, one function's output may be a) an input to another function, b) a resource, c) a pre-condition, or d) a control or time constraint is imposed. The effect is an overall explanation of how in the accident scenario the functions were related or connected, and thus a description of how functional variability propagated through the system. The ties usually define where the variability of one feature may have an effect or how it can spread. A resonance effect may be produced by several such occurrences and propagations of variability: although the variability of each feature may be below the usual detection threshold, they can become a 'signal' in combination, so this is a risk. A visualisation of how the functions are related can help this step. In tracing functional dependencies, this form of visualisation can be useful, but the study should nevertheless be based on the function description rather than on the graphical representation.

**Step 4.** Identify variability barriers (damping factors) and determine the performance monitoring necessary. Barriers are the means of avoiding the occurrence of an unwanted event, or of defending against the effects of an

unwanted event. It is possible to define barriers in terms of barrier structures (the barrier's organisational and/or physical structure) and barrier functions (the way the barrier achieves its purpose). The four fundamental systems of barriers are:

- Physical barrier systems that block mass, resources, or information movement or transportation;
- Functional barrier systems that set up pre-conditions that must be satisfied before an action can be taken (by a person and/or machine);
- Symbolic barrier systems Symbolic barrier structures that are signs of physically existing limits on action; and
- Incorporeal barrier system that signify action constraints that are not physically present.

In addition to the barrier recommendations, the FRAM analysis can provide the basis for the issuance of performance monitoring recommendations in order to identify excessive variability. It is possible to establish performance indicators both for functions and for couplings between indicators.

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## 14.6 TYPES OF ANALYSIS OF ACCIDENT

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There are several types of investigations and analysis of accident. Some of them are as follows:

### 1) Failure Mode and Effect Analysis

In the failure mode and effect analysis failure or malfunction of each component, the mode of failure and its ultimate effect from the overall performance is evaluated. This is a straight forward method provided that the analyst has a thorough knowledge of the system.

Its drawback is that it considers only one failure is that it considers only one failure at a time and some probability may be overlooked.

It could be very useful for investigating where large complex, inter-related machines and processes are involved.

### 2) Fault-Tree Analysis

In the fault-tree analysis method, an undesired event is selected and all the possible happenings that can contribute to the event diagrammed in the form of a tree. The branches of the tree are continued until independent events are reached

Probabilities determined for the independent events can be computed. This is a very powerful analysis technique but requires complex mathematical calculation. This method yields good results if management procedure, communication and relationship to accident is of interest.

### 3) Theory of Human Error

This is a American technique used to quantitatively evaluate the contribution of human error in degrading the work situation and be called human

component. It can be combined with any of the two earlier method described for computation of study.

#### **4) Cost Effectiveness**

In the cost effectiveness method, cost of system change made to increase to safety are compared with either the decreased cost of lower serious failure or increased effectiveness of the system to perform his task to determine relative value of these change. This method helps in selecting the most economical method yielding same result.

#### **5) Statistical Method**

Statistical method is a conventional one. The defects are grouped together based on four M's (Man, Machine, Material, Management) of three E's (Engineering, Enforcement, Education).

Accidents are distributed among these categories and problem components identified preventive action.

#### **6) Critical Incidence Method**

These techniques consist of selecting at random a number of workers from major plant / department and integrating them to seek errors they might have committed or observed for similar jobs in the past.

The incident described are classified into various categories and accident problem are seen

These techniques can identify mechanical and human errors which may have contributed to an accident or have accident potential.

#### **7) System Safety**

System safety method believes in a complete view of an inter-relationship of various events that can lead to an accident.

It can widely cover including machine tools, materials, environmental factor, people, documents, operating instructions, training manuals etc.

As accident is rarely caused by one factor, the system safety method may identify more than one problem areas where counter measures are necessary.

It has the advantage of application to a new process and plant and can be applied before accident can occur. It transforms the safety professional and art of science by codifying knowledge.

#### **SAQ 2**

- a) Write down the different types of analysis of accident.
- b) Write short notes on:
  - (i) Failure mode & effect analysis
  - (ii) Fault-tree analysis
  - (iii) Critical incidence Technique
  - (iv) System safety

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## 14.7 BASIC COMPONENTS OF AN ACCIDENT INVESTIGATION

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The basic components of an accident investigation process are:

- The accident sequence.
- Equipment factors analysis.
- Environmental factors analysis.
- The nature and complexity of the accident determines the extent to which these components are evaluated.
- Accident Sequence

The series of accidents consists of five components and is only defined on the basis of the evidence decided during the investigation. The five ingredients are:

- Events occurring before: Pre-accident events Determine the series of events leading to the accident in order to address the questions: who, what, where, where, and how. Identify any contributing variables, such as urgency, weather, state of equipment, or terrain. If there was a fire involved, decide where, where, and how the fire began. Determine the spread of flames and whether efforts have been made to extinguish the fire.
- The accident sequence: Begin with the initiating incident (for example, the truck tyre blew out or the tail rotor hit a snag) and proceed until the sequence reaches a reasonable endpoint.
- Events occurring after the accident: Identify the sequence of events (such as search and rescue or medical efforts) that occurred after the crash, how the accident was first identified, and the post-accident staff and equipment locations. Note any crash site disruptions and protection or preservation steps taken, as well as any accidents and fatalities or contributing factors, such as rescue and emergency response, due to incidents that occurred after the accident.
- Injuries: Document any wounds. Identify all medical facilities that provided care, record the patients' condition, and, if necessary, summarise autopsy reports.
- Damage: Estimate the cost of damage to the equipment or properties and describe the damage as minor, significant, disabled, or repairable.

### Human Factors Accident and Incident Analysis

In most injuries, human factors play a significant role. The human factors that lead to an accident need to be able to be identified by investigators.

Successful intervention and preventive strategies may benefit from careful review.

**Qualifications and Training:** Determine the skills and training of persons (the vehicle operator, passengers, and supervisor) directly involved in the accident. Identify some contributing factors, such as the absence of operator certification or inadequate instruction.

**Duties:** Identify the roles of those directly involved in the crash, such as primary and additional tasks, and arrangements for work and rest. Notice any contributing variables, such as the exhaustion of employees. Perform an overview of work / rest covering at least 72 hours before the accident. Require an analysis of time and attendance records as well as feedback on tasks performed and real time worked (may not always fit reported time), off-duty activities, and periods of sleep duration from relevant supervisors.

**Management:** Determine the company, oversight, and external supervision of people directly involved in the accident. Identify any contributing variables, such as a failure by the supervisor or organisation to prioritise protection.

**Compliance:** Note any deviations observed from procedures, systems, methods, and requirements of contracts. Study the JHA, protective devices, and other things related to the investigation into the accident.

**Documents:** Identify if the accident-related individuals were present, readily available, and appropriately used by directives, operating manuals, and contracts. Evaluate reports that are unique to the accident, such as inspections, logs of dispatch and facilities, records of time and attendance, safety plans, and, if necessary, incident command system forms.

**Communications:** Before, during, and after the accident, describe the type of communication used. Identify any communications-related contributing factors, such as radio coverage or defective devices.

**Services:** Determine if the accident contributed to contracted facilities, such as road guards, traffic signals or dispatch procedures.

**Risk Management:** Determine whether a JHA or other risk analysis for the workplace has been established. Establish the role played in the success of the job project or operation through risk analysis. Determine if, before work started, a tailgate safety session was conducted and recorded.

### **Equipment Factors Analysis**

**Systems:** Determine the equipment involved in the accident and its suitability for the job project or operation to be done. Include any applicable manuals for the operator, maintenance records, inspections, and maintenance staff approvals.

Survival: Review the vehicle, device or equipment 's capacity and suitability to conduct the work project or operation and the structural integrity of the occupant compartment.

Impact conditions and crash (dynamic) forces.

Restraint and rollover protection systems. Were such systems installed? Were they used?

Personal protective clothing and safety equipment.

Backup and emergency systems.

Safety design.

### **Laboratory or Tear down Analysis**

Review the results of any equipment component analyses. Special studies or tests should be conducted by another agency or private laboratory.

### **Environmental Factors Analysis**

Weather: Before, after and after the crash, check the environmental conditions. Identify any variables that relate, such as rainfall , temperature, illumination, and visibility.

Physical Environment. Describe the accident scene entirely. Determine if the scene has been preserved. Notice the landscape at the site of the crash.

Provide a general area chart, a site-specific position map, landscape feature profiles, accident site diagrams and sketches, and diagrams of all other related items. From a control point that has some permanency, take all measurements. During return trips to the site, measurements can be performed from the control point. Identify any factors that contribute, such as elevation, vegetation, slope, accessibility, dust, and smoke. The components of Accident Chain for Accident Analysis are given in Table14.1.

**Table 14.1: Components of Accident Chain for Accident Analysis is given in table14.1.**

| <b>The accident</b>       | <b>Result of Accident</b>               | <b>Immediate Causes of Accident</b>     |
|---------------------------|-----------------------------------------|-----------------------------------------|
| • Fall                    | • Annoyance                             | Unsafe acts<br>and<br>Unsafe Conditions |
| • Slip                    | • Spoilage                              |                                         |
| • Slide                   | • Property Damage                       |                                         |
| • Collision               | • Minor Injuries                        |                                         |
| • Being caught in between | • Disabling injuries<br>• Fatality      |                                         |
| • Eruption/Explosion      | • Production Delay<br>• Reduced quality |                                         |
| • Burn/Fire               | • Cost of re-processing                 |                                         |

## **Contributing causes of accidents**

### **a) Supervisory Performance**

- Safety instruction inadequate
- Safety rules not enforced
- Safety not planned as part of job
- Infrequent employee safety contact
- Hazards not correcting
- Safety devices not provided

### **b) Mental condition of a person**

- Lack of safety awareness
- Lack of co-ordination
- Improper attitude
- Slow mental reaction (Fear of accident)
- Inattention
- Lack of emotional stability
- Nervousness
- Temperamentation etc

### **c) Physical Condition of a person**

- Extreme fatigue
- Deafness
- Poor eye sight
- Lack of qualification / knowledge for the job
- Hearing condition of environment
- Any other handicap problem

(Natural rhythms) etc.

### **SAQ 3**

- Explain industrial hazard ?
- What is safety management ?
- Write down the components of accident chain & their results & causes.
- What are the contributing causes of accidents? Explain.

## 14.8 CASE - FIRE AT JAIPUR OIL DEPOT IN 2009

It was the evening of October 29, '2009, when the fire at Jaipur oil depot of Indian Oil Corporation (IOC) broke out which leading to death of more than 12 people and wounding at least 200. A glimpse of the same is shown in Figure 14.1.



Figure: 14.1: Jaipur Oil Depot Fire

It was the evening of October 29, '2009, when the fire at Jaipur oil depot of Indian Oil Corporation (IOC) broke out which leading to death of more than 12 people and wounding at least 200.

The fire started in an oil depot's giant tank holding 280,000 cubic feet of oil. Though, the depot area is on the outskirts, about 16 kilometers south of the city of Jaipur, the blaze was so intense and uncontrolled that it continued for more than a week. About 5 lacs people from the nearby area were evacuated after the incident due to intense fire and smoke. Petrol was being transferred from the IOC's depot to a pipeline at the time of accident.

More than forty employees of IOC were present at the terminal at the time of explosion and fire. Even the met department recorded a tremor of magnitude 2.3 on the Richter scale at the same time when the first explosion took place at 7.35 P.M. Such a huge explosion resulted in shattering of window's glass within the periphery of around 3 kilometers from the site of accident. The cost of accident was huge. It resulted in not only the human life and injury but also led to loss of business, infrastructure, migration of people and huge impact on environment. Even the common people in close vicinity faced difficulty in breathing for few hours due to presence of vapors of petrol in the atmosphere and smoke. Some of the hotels around the site had to ask the guests to vacate to avoid any tragedy.

The accident showed how a small error could lead to catastrophic event and the unpreparedness to deal with such a situation. Adjacent to the terminal wall was the workshop of morani motors (p) limited. Where, as per eyewitnesses the car parked on the roof top were thrown up in air to about 10 feet and around 35 new Hyundai brand cars were completely damaged.

The local police, administration and fire emergency services reached the spot. Even police chief, district collector and general manager of IOC reached the spot but had no concrete plans and capabilities to deal with such a situation. Around half past six the staff in the terminal had contained the leak and flow of petrol panicked and reported the matter to local police station. The nearby industries which were running second shifts were cautioned to vacate the area. At around 7.35 pm, a huge ball of fire with loud explosion broke out engulfing the leaking petrol tank and other nearby petrol tanks with continuous fire with flames rising 30-35 m and visible from a 30 km radius. The traffic on adjacent national highway no 12 was stopped leading to a 20 km long traffic jam. On 30<sup>th</sup> October Army as well as experts from Mumbai were called and deployed and then only the situation contained. The district administration decided to disconnect electricity and evacuate the nearby areas to limit the damage in case of further worsening of situation. The fire still raged on 31 October 2009 in the Indian oil corporation depot at Jaipur after a defective pipeline leak that set fire to 50,000 kilolitres of diesel and petrol out of the storage tanks at the IOC depot. By then the accident had already claimed 11 lives and seriously injured more than 150. This accident exposed the unpreparedness of IOC and district administration to deal with such a calamity. They had no plan to control the situation in such an event. The equipments available with the local fire control department and the capability to deal with such disasters was much below the standards. They remained onlookers and no efforts were made to breach the terminal wall to get closer to kerosene and diesel tanks to cool them with water jets. Such a disaster in oil installation is an eye opener for all the oil companies as well as for the companies dealing with explosive chemicals and gases for various industrial operations to keep and maintain adequate fire & safety facilities. They should adhere to the safe practices and also keep in place a firm inspection procedure for safety audit.

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## **14.9 LET US SUM UP**

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Thus, all accidents are caused. No accidents occurs automatically, instead certain factors cause accidents. It has been noticed that an accident does not have a single cause but a multiplicity of causes which are often closely related. They are the result of human error which involves unsafe behavior or an unsafe condition or a combination of both.

There are some other causes which arise out of unsafe situational and climatic conditions and variations. They may include excessive noise, very high temperature, humid conditions, bad working conditions, unhealthy environment, slippery floors, excessive glare, dust and fume, arrogant behavior of supervisor etc.

**Discovering “what happened” and “why” are the objectives of an accident investigation.**

Supervisor and employee work together to identify causes and remedial measures / actions. Everyone involved should know that an accident investigation is not a “fault finding” or “finger pointing” exercise. Meaningful involvement is essential. Employee can provide valuable suggestions when they are called in the investigation process.

Industrial accidents cause losses to the employees and organizations as well. Accidents causing losses to the industrial establishments need to be avoided. Adequate safety measures can avoid accidents.

Industrial safety refers to protection against accidents occurring in the industrial establishments. Hence safety analysis and accident investigations plays a very important role.

It helps to understand how a situation can be made safer. It also helps to develop safety improvements for safety management within the organization. In totality it can be said that safety analysis is a procedure for analyzing total systems to identify and evaluate hazards and safety characteristics.

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## 14.10 KEYWORDS

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**Accident:** An accident is an event that causes unintentional damage or injury.

**Incident:** An incident (near-accident) is an event that almost causes unintentional damage or injury.

**Hazard:** The term hazard is often used to denote a possible source or cause of an accident.

**Risk:** It can be defined as the possibility of an undesired consequence.

**Accident investigation:** An accident investigation is the collection and examination of facts related to a specific occurred accident.

**Risk management:** Risk management consists in coordinated activities to direct and control an organization with regard to risk.

**Safety management:** Safety management is a way of managing hazards and risks in an organization.

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## 14.11 ANSWERS TO SAQs

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For all SAQs, please refer to the relevant text in the unit.

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## 14.12 REFERENCES AND FURTHER READINGS

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## UNIT 15 COSTS OF ACCIDENTS

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### Structure

- 15.1 Introduction
  - Objectives
- 15.2 Lessons from Past on Major Industrial Accidents and their Cost
- 15.3 Accident Costs
- 15.4 Types of Costs
  - 15.4.1 Financial Cost
  - 15.4.2 Human cost
- 15.5 Tools for Accident Cost Analysis
  - 15.5.1 Insurance Based
  - 15.5.2 Activity Based Approach
  - 15.5.3 Labour Capacity Based Approach
- 15.6 Let Us Sum Up
- 15.7 Key Words
- 15.8 Answers to SAQs
- 15.9 References and Future Studies

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### 15.1 INTRODUCTION

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In the previous units we have discussed about accidents, types of accidents and various theories related to industrial accidents. This unit deals with the economic aspects of accident. The cost of industrial accidents is so high, that it has become essential for all the employers as well as the government organizations linked to the industries to find the reason of such industrial accidents and to minimize them to near zero.

The cost of an accident is very heavy both for the employee as well as the employer. The accidents are associated with the definite costs such as direct and indirect costs. The direct cost implies the measurable damage to the system and the employee whereas, the indirect costs are intangible but nevertheless real costs.

In most of the organizations, data related to cost of accident is limited to the fees and subscriptions payable to the insurance companies for occupational accidents and diseases, any damage not covered in insurance and fine payable to the government. Whereas, the cost assessment involved in preventing the occupational hazard, the cost of production loss and the downtime cost of other workers and managers are very less frequently conducted by the employers. Moreover, the cost associated with the damage to the company image which leads to the loss of market is also generally not calculated by the employers.

Thus, an incomplete or incorrect estimation of the cost of occupational health and safety may lead to huge losses to the organization and diminishing the market image of the company. Amount spent on safety should be considered as an investment and not an expenditure. Good safety practices lead to lower cost of accidents, improves morale of the employees and improves the market image.

### **Objectives**

After studying this unit, you should be able to

- describe the cost of industrial accident,
- compare the different costs involved in industrial accident,
- discuss the different tools for accident cost analysis,
- calculate the cost of industrial accident.

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## **15.2 LESSONS FROM PAST ON MAJOR INDUSTRIAL ACCIDENTS AND THEIR COST**

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Some of the examples from the past are mentioned in this section to have an idea of how much an accident can actually cost to a company.

### **1) Deepwater Horizon Oil spill, Gulf of Mexico 2010**

**Damage:** 11 workers died in an explosion and fire. This accident is the largest oil spill accident in the history of petroleum industry. An estimated 4.9 million barrels of oil is spilled into the gulf waters for over 87 days. It continues to cause extensive damage to wildlife habitats, fishing and tourism in the region.

**Cost:** The spill has cost BP \$42.7 billion so far. Possible cash penalties under America's Clean Water Act could cost an additional \$20 billion.

### **2) Dhaka Building Collapse, Bangladesh 2013**

**Damage:** 1,126 people are believed to have died, while 2,000 more were injured in what was the worst garment factory accident ever. The eight-storeyed building housed factories of high street brands, such as Benetton. Primark, Bom Marche and Mango, shops and a bank.

**Compensation:** Although families have received short-term compensations from the Bangladesh government and Primark, a final compensation remains elusive. After talks between clothing brands and pressure groups in Geneva last September proved futile, four global retailers have set up a \$40 million compensation fund, which is open to contributions from international donors.

### **3) Chernobyl Nuclear Disaster, Ukraine 1986**

**Damage:** Between 1986 and 2000, more than 350,000 people were evacuated from Belarus, Russia and Ukraine. Soviet Union claimed the accident killed 31 people. Continuing analysis come up with numbers that vary enormously; they estimate 4,000 deaths. This does not include about 50,000 people living

with cancer, of whom 25,000 are expected to die. Flora and fauna in the area was heavily affected, with animals and trees either dying or not reproducing. The effects are expected to remain for the next 100 years, with declining intensity.

**Cost:** The Soviet government almost went bankrupt, spending 18 billion roubles (worth \$18 billion at that time) in containment and decontamination; for Belarus, the total over 30 years is estimated at \$235 billion; 5-7 percent of Ukraine's government spend is related to the accident.

#### SAQ 1

- a) What do you understand by cost of accident?
- b) Gives some examples of industrial accidents leading to huge loss for the organization.

### 15.3 ACCIDENT COSTS

The cost associated with the occupational accident is defined as the “effects on the costs of a company that would not have been borne if the injury/accident had not taken place”. The calculation of the cost of accident is very complicated and it involves various difficulties and methodological problems. The primary difficulties is to identify the consequences which are driven from an occupational accident. Most of these consequences are unknown due to the casual relationship between the accident and the consequences. Also, all the consequences does not occur at the same time or place. Other difficulties includes the availability of limited time and resources, apparent intricacy of the evaluation, biased accounting system, ethics, incapable or completely absent occupational health and safety department. Various previous studies provides the detailed methodologies for the classification and evaluation of accident cost. However, most of the studies only consider a subset of consequences. One such classification of the accident cost is the external and internal cost. Here, the internal cost refers to the loss of productivity and the time spent on the investigation. Whereas, the external cost involves the hospitalization and medication charges incurred for the treatment of victim which are not covered under the insurance, loss of time and resources on the works household nursing and recuperation. Another classification is the fixed and variable cost which shows that the all cost are not visible. This classification shows that the cost of accident depends largely on the incidence of work accident and work related health. The last classification deals with the tangible and in-tangible cost of accident. These cost referees the costs which cannot be expressed in monetary values.

In parallel, several tools are available for the cost analysis of accidents in the industries. These tools are classified in three different approaches:

- i) Insurance based approach,

- ii) Activity based approach, and
- iii) Labour capacity based approach

The insurance based approach involves the cost analysis only from the perspective of insurance. It primarily focus on the costs which are to be reimbursed and the costs which are to be borne by the organization. The costing models which are developed considering this peculiar approach works on the basis of information provided by the worker's compensation insurance and generally available in the company's accounting system. Thus in this approach the data availability is not an issue. However, an in-depth knowledge of insurance and its complications are required which is very difficult to address. Though, the insurance based approach is very vital, it does not provide a method which can be implemented practically for the companies. Also, this method incompletely categorizes the cost occurred to the organization in case of an accident. The primary objective of this approach is to provide the detailed cost breakup of the accident at work and work related ill health by evaluating the ratio between the various cost categories.

The activity based approach involves the development of cost analysis tools that can be implemented practically in the organization. It focuses on supporting the decision making abilities by taking into consideration the cost involves in health and safety measures. This approach works on the basis of documentation of all the consequences generated due to an accident and, consequently, the cost evaluation of these consequences. The activity based approach provides a tailor-made analysis and thus, enhances the transparency the cost evaluation of the accident.

The labour capacity based approach evaluates the consequences of absenteeism instead of economical quantification of the accident. This type of analysis can be conducted through various established methodologies. The methodologies ranging from evaluating the cost of absence and interrupted production, to calculating the costs per uninterrupted working hour.

Among the above mentioned three approaches, the labour capacity based approach is the scantily reported in the literature owing to its objective which focuses on the analysis of productivity instead of the investigation of the accident. Thus, the activity based approach is most suitable of the identification of all the consequences and the costs associated with these accidents which further assists in the decision making.

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## **15.4 TYPE OF COSTS**

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The overall cost of an accident in the industry is broadly characterized according to the nature of accident and its consequences. The flow chart depicted in Figure 15.1 illustrates the various costs involved in the occupational accidents.

## 15.4.1 Financial Cost

### a) Direct Cost

- i) **Cost of damage:** It includes the cost associated with the damage to the product, equipment, machinery and inventory. Moreover, the cost involved in cleaning and returning the working area back to the functioning is also considered as the cost of damage.
- ii) **Medical cost:** It refers to the immediate medical expenses incurred by the company after the accident. It involves the immediate evacuation to the hospital, medical expenses at the accident site, hospitalization.

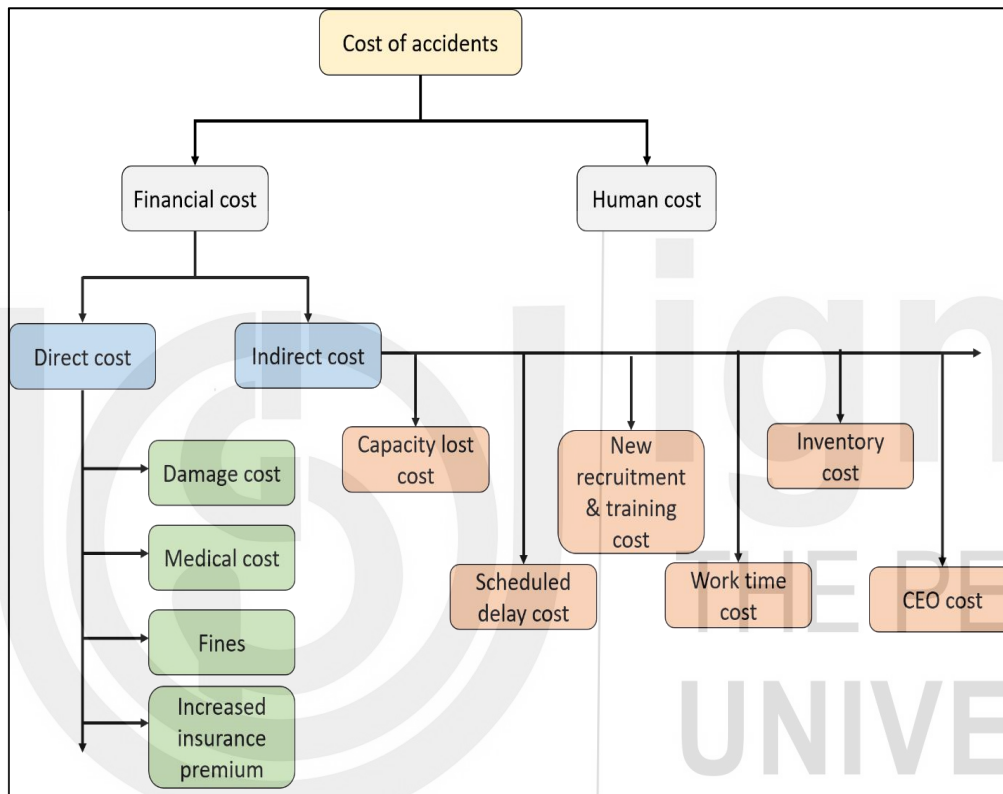


Figure 15.1: Flow Chart for Various Costs Involved in the Occupational Accidents.

- iii) **Cost of fines:** In case, the company has violated any safety procedures which leads to an accident, the heavy fines are imposed on it by the authorities.
- iv) **Insurance cost:** Increase in the insurance premium after an accident. The annual insurance premium liable for the company depends upon the estimate of absence leaves, hospitalization periods, severity of accident, potential lawsuits and the financial damage of equipment, commodities and infrastructure. The premium amount of a particular year is generally depends upon the events of the previous year. Thus an accident leads to the drastic increase in the premium amount for the coming years.

## b) Indirect Cost

All costs for accidents have been estimated for many years; it would seem that the estimates of indirect costs have been underestimated for many serious accidents. Traditionally the indirect costs for accidents have been estimated as follows:

- i) **Cost of capacity lost: The costs resulting from production loss.** An accident can retard the production or even can stop it for a particular period of time, for example, emigration of the injured workers and damage to the equipment which should be handled immediately (like fire). Also, an accident may result in a new bottleneck causing production processes to slow down and imposed additional costs.
- ii) **Cost of scheduled delays:** When an accident occurs, slowdown in production will affect the time table schedule and causing damages to the client. Clients can cancel the contract or demand a lower price. There may be a solution that the company will create the absented product by contractor that will help the company to handle the schedule.
- iii) **Cost of recruitment & training:** If a worker is being injured in an accident, he/she is required to be replaced with other worker. It involves the cost of recruitment and training for the new workers who replace the injured worker. Training on new worker also contributes to time lost in addition to the cost of training.
- iv) **Cost of work time:** The work managers invest in investigating the accident. Work time is also dedicated to instruction of the simple workers. Also the additional work hours that needed to replace the injured worker (it depends on the policy of the company if there are recruiting new workers or letting the senior to work extra hours).
- v) **Cost of Inventory:** It is common to have a situation of bottleneck after the accident. This bottleneck leads to increase in the level of inventory. Inventory shares a huge cost of the plant operation and leads to blockage of huge money in inventory. Thus, in this situation it is required to find a solution to manage the inventory thus avoiding the additional expenses on storage.
- vi) **Cost of Managers and Executives:** Executives and managers are highly paid in any organizations. Accident in organization leads to formation of various committees and sometime legal actions against the plant. In these type of situations, executives and managers are required to spend time in these activities leading to ineffective utilization of their time. It again adds to the cost of accidents.

### SAQ 2

- a) What are different types of cost associated with industrial accident?
- b) Illustrate various costs of occupation accidents with a flow chart.

The various approach for accident cost analysis are as follows:

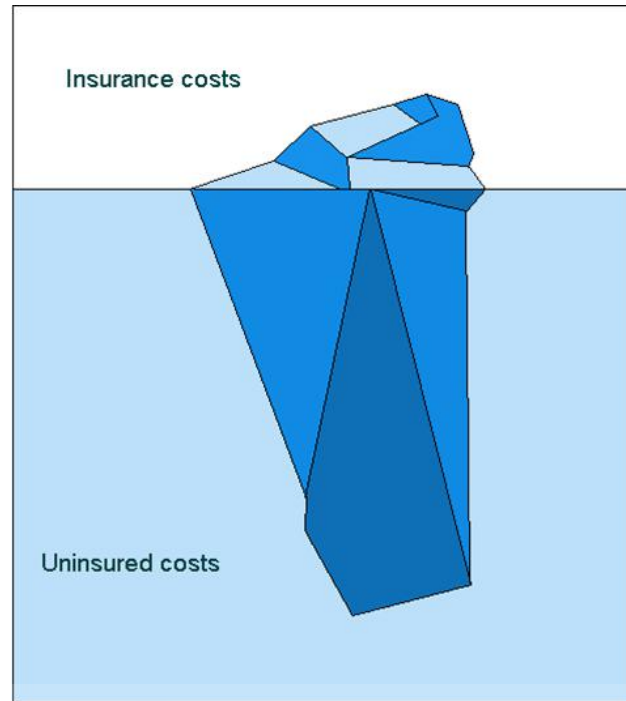
### 15.5.1 Insurance Based Approach

The origins of the insurance based approach can be traced back to work done by Herbert Heinrich in the 1920s. Heinrich documented that the costs of occupational accidents in American companies were substantial and that many costs were hidden from the view of management (Heinrich 1959). Heinrich used data from insurance cases and divided costs into direct and indirect costs where the main classification criterion was whether these costs were refunded by the insurance. The costs that were not refunded by the insurance were termed indirect or hidden costs and were, according to Heinrich's study of selected occupational accidents, approximately 75% of the total costs of an average occupational accident. To express this distinction Heinrich developed the iceberg metaphor where the larger hidden part of the iceberg represented the indirect costs as depicted in Figure 15.2. One example of direct costs is e.g. sick pay and examples of indirect costs are e.g. the non-productive time of colleagues, administrative costs, production setbacks, replacement hiring costs, fines and investments in extra safety measures (Andreoni 1986).

The common theme in these studies is the fact that they try in some way to distinguish between hidden and visible costs and usually apply the insurance criteria to do so. That is to say costs are analyzed in an insurance perspective and there is a lot of weight on what costs are refunded and what costs are not. This in turn often implies that the cost categories used are defined beforehand and often require some knowledge of insurance issues before they can be applied consistently across industries and companies. In some of the studies cited above this has meant that the analysis themselves is carried out by academics or outside consultants rather than by e.g. safety managers.

### 15.5.2 Activity Based Approach

In decades of 70's and 80's an innovative approach emerged for the study of consequences of an industrial accident. Concept of controllable and non-controllable costs in the context of costs of occupational accidents was introduced by Laufer et al. in 1987. This concept becomes a milestone for investigating the consequences from the management oriented approach.



**Figure 15.2: Accident Costs Iceberg**

The activity based approach focuses on the management and the efficiency of the management to implement health and safety costs in their decision making policy to avoid these unnecessary expenditure. Consequently, the management is actively involved in these studies and accelerates the development of tools and techniques that can be implemented by the management. Therefore, several studies on the activity based consequences are conducted in close cooperation with the management. In the consequence based studies, the insurance does not plays a crucial role and thus the involvement of management is very often mentioned as being more prominent.

The key attraction of this approach is the link between the accident and the consequences which are then valued in economic terms. In other words, the activity based approach focuses on the activities that happen after the accident and the economic impact of these activities. Insurance as normally does not play any significant role in the activity based approach due to which it further leads to an another difference between the visible and hidden costs. Activity based studies often does not consider the insurance criteria but base their distinction between hidden and visible costs on whether management has these costs readily available from the accounting systems of the company.

#### **(i) The Accident Consequence Tree (ACT) method**

The objective of the Act method is to focuses the impact of consequences on the society, a company and the injured employees of the organization. The method follows a same procedure for all the three receivers i.e. the company, the society and the injured employees. The procedure to apply an ACT method consist of:

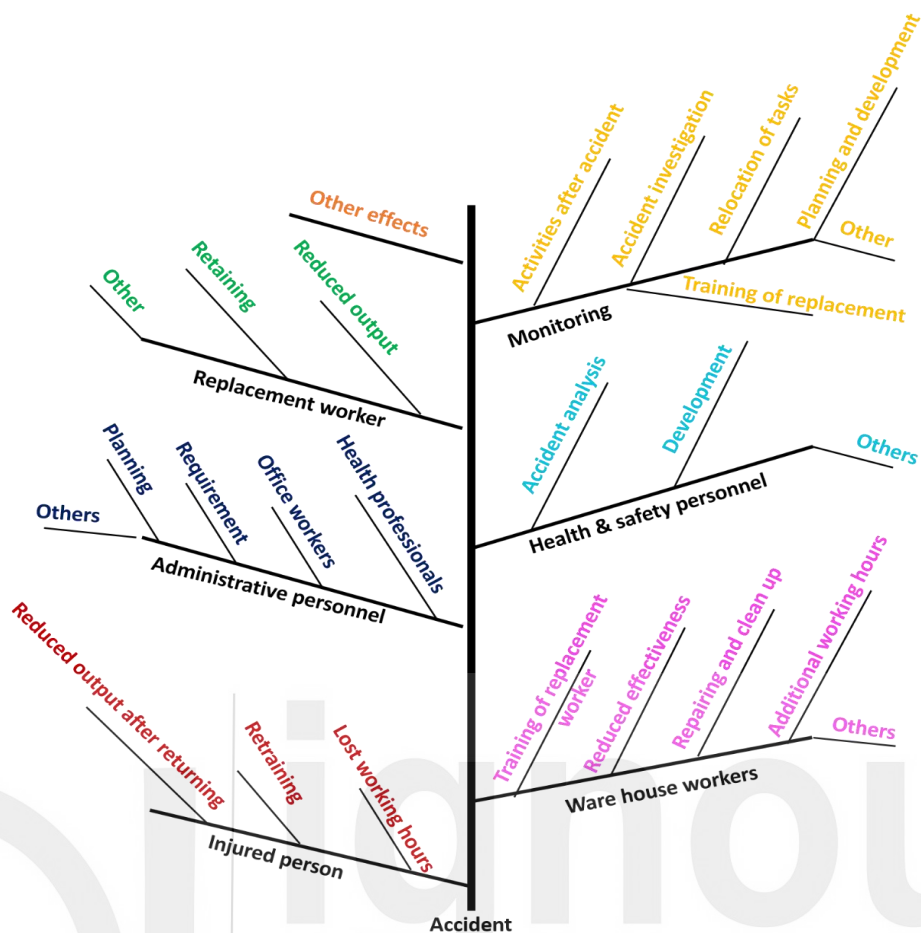
1. Occurrence of an event
2. Identification and registration of the consequences and activities in real time
3. The format in which the consequences are quantified are such as number of hours, number of visits, quantity etc.
4. Allocation of unit cost to the identified quantities.
5. Calculation of the cost.

The social cost of an accident on the society is a direct cost. It includes the ambulance expenditures, consultation cost etc. The cost related to an injured worker involves the reduction in family overall income and an additional medical expenses which are not covered under the insurance such as juice, special diet, nursing etc.

The social costs are the direct costs that result from the accidents i.e. the costs of an ambulance trip, the costs of the doctors involved etc. The costs related to the injured worker include the reductions in income and any extra costs incurred by the worker or his/her family due to the accidents. There are 6 main categories used in the ACT method to classify consequences and cost on the company due to an occupational accident:

1. Loss in the working time.
2. Loss of inventory and products
3. Loss of heavy machines and infrastructure
4. Diverse short term costs such as costs of transport, consultants and fines
5. Loss of contracts and drop in the price.
6. Income such as reimbursement form insurance provider
7. Increased insurance premium

Figure 15.3 shows an example of an accident tree based on Aaltonen (1996).



15.3: An Example of an Accident Tree Based on Aaltonen (1996).

## (ii) The Riel and Imbeau ABC method

Riel and Imbeau (1995 a, b, c, d; 1996; 1998) base their method (here called the R&I method) on the activity approach but in a somewhat different manner than the ACT method described above. Their method focuses on calculating accident costs with the purpose of using these costs as an allocation base for insurance costs. This is then used for evaluating the effects of ergonomic investments on insurance costs. This method thus combines to a degree both a consequence and an effect study. One issue is however, that the author's definition of "ergonomics" is rather broad and includes both workplace design as well as accident prevention measures.

Riel and Imbeau develop their method on the basis of Activity Based Costing (Kaplan & Norton 1997). Activity based costing (ABC) reflects the same approach as the ACT method in that it values in financial terms the activities that are generated by an event. This application defines the event as a cost object and the measurements concern the resources that are consumed by the activities that are related to this event.

**The method is based on the following stages:**

1. Evaluation of H&S costs. An important criterion is that the cost generated could have been avoided if the accident did not occur. These costs are evaluated through interviews and analysis of cost registrations.
2. Identification of cost behavior – i.e. the cost drivers and the causal relationships with the costs in question. The R&I method distinguishes between three drivers which are resource drivers, activity drivers and cost drivers.
3. Cost allocation where the costs identified in stage 2 are used for allocating a cost pool which in the case of the R&I method are insurance costs but could in principle be any other type of cost provided there exists a causal relationship.
4. Cash flow calculations for the investment or initiative that is to be evaluated
5. Investment evaluation involving calculation of e.g. Return on Investment or Internal Rate of Return.
6. Investment evaluation which is carried out after the investment has been carried out and looks at whether the investment has been successful in relation to the chosen criteria.

When evaluating the costs of occupational accidents Riel & Imbeau use a framework where costs of accidents are likened to production disturbances. The costs of these disturbances are classified into:

1. Over consumption of materials and assets – i.e. use of materials etc. which would not have taken place if the disturbance had not happened.
2. Over consumption of time – i.e. payment for employee time conducting activities that would not have been conducted if the disturbance had not taken place
3. Lost working hours – i.e. payment for working hours where no activities are carried out
4. Lost production – i.e. lost raw materials, capacity, products etc. due to the disturbance.

The R&I method has been tested in an aerospace company in Canada where insurance costs were allocated between 6 departments on the basis of accident costs. Previously these insurance costs had not been allocated but accounted for as factory overhead. By allocating these costs on the basis of where the cost claims were generated the company was able to get a more accurate picture of the costs structures involved.

### **(iii) Systematic Accident Cost Analysis (SACA) Method**

The SACA method was developed by the Aarhus School of Business and PricewaterhouseCoopers in Denmark (see Rikhardsson et al. 2002; Rikhardsson&Impgaard 2004; Rikhardsson 2004).

Being a consequence based methodology like the ACT and the R&I method the SACA method focuses on the consequences of occupational accidents and the costs of these consequences. The main procedure is first identifying the activities generated by the occupational accident being analysed and the next step is calculating the costs of these to the company. There is no allocation of costs like in the R&I method and there is solely focus on company costs and not on societies costs or the costs to the injured employee like in the ACT method.

The SACA method is based on a number of forms and checklists which are intended for guidance only. The main aim of the method is to provide managers with a tool for identifying accidents costs without them having to have a lot of experience in accounting or financial analysis to be able to do so. The method is based on interviews and workshops and is not intended to be used in real time.

The costs of occupational accidents are classified into six overall categories:

1. Costs due to the absence of the injured employee: Includes e.g. payment of sick pay and payment of supplementary sick pay.
2. Communication costs: Includes e.g. formal communication to employees, staff, and general management as well as informal communication between employees.
3. Administration costs: Includes payroll administration, administration regarding health and safety regulations and reporting requirements, follow-up activities and meetings.
4. Costs of prevention initiatives: Includes e.g. purchase of machine components and training initiatives.
5. Operation disturbance costs: Includes e.g. training of replacements, revenue loss, co-workers overtime, and production reductions.
6. Other costs: Includes costs such as e.g. fines and gifts to injured employee.

The actual costs measured within these categories are grouped into four categories:

1. Time: Hours used by employees and management on activities related to the accident as well as hours for which the company pays wages without getting any work effort in return including standstill periods in production and employee sick pay.
2. Materials and components: Costs of any materials and components acquired or lost due to the accident such as spare part for machines, replacement for damaged materials, and value of products not produced.

3. External services: Costs of external services bought due to the accident such as temporary replacements, consultants and legal support
4. Other costs: Costs of other activities more infrequently incurred by the company such as fines and rehabilitation.

#### iv) The Health & Safety Executive (HSE) method

The method is developed by the Health & Safety Executive in the UK in the early 1990s (HMSO 1993). As such it differs from the three methods above as is more focused on insurance costs and uses the insurance criterion to distinguish between hidden and visible costs. Furthermore, it not only includes H&S costs but focuses on material damage costs as well even though no injury is associated with this damage. The cost definitions of the HSE method are shown in Figure 15.2. It distinguishes between direct and indirect costs which may or may not be refunded by the company insurance.

Furthermore, the method is applied in real time like the ACT method.

The method also distinguished between what it terms financial costs and opportunity costs. Financial costs are defined as the costs incurred due to the necessary activities resulting from the accident and opportunity costs are costs for which the company gets no value in return. This is not quite the usual accounting definition of the term opportunity costs and the term non-value added costs might have been more appropriate. The method also stresses that only costs which could have been avoided if the accident had not happened should be registered.

#### 15.5.3 Labour Capacity Based Approach

This approach is based on assessing the consequences of absenteeism. It does not quantify the work accidents in terms of monetary values. This type of analysis can be seen in several methodologies. Some uses the costs of absenteeism as a calculation; some consider the lost production and few other utilized the uninterrupted working hour for calculation analysis. The labour capacity-based approach has been the least developed in the literature because out of the three classes of methods because it mainly focuses on productivity level analysis and not on an investigation of work accidents.

#### Example 1:

Cost of an accident. A worker got injured on his arm while working in the steel melting shop.

| Direct costs                        | In INR |
|-------------------------------------|--------|
| Compensation paid for injuries      | 50000  |
| Hospital/Medical expenses on injury | 75000  |

|                                                                                                                    |               |
|--------------------------------------------------------------------------------------------------------------------|---------------|
| Total compensation cost                                                                                            | 125000        |
| Uncompensated wage loss                                                                                            | 25000         |
| Total direct cost                                                                                                  | 150000        |
| <b>Indirect cost</b>                                                                                               |               |
| Material wasted and labour for disposing it                                                                        | 20000         |
| Pay to injured worker while at home                                                                                | 20000         |
| Loss of manhour of fellow workmen looking at the accident (not working )                                           | 10000         |
| Supervisor's time lost in investigating and recording incident                                                     | 20000         |
| Downtime of the system                                                                                             | 150000        |
| Reduced production rate of other employees                                                                         | 15000         |
| Total indirect cost                                                                                                | 235000        |
| Total cost of accident [Excluding overhead charges which may rise the total cost of accident by as much as 30-40%] | <b>385000</b> |

### SAQ 3

- Explain in detail different tools for accident cost analysis
- Illustrate accident tree based on Aaltonen theory.
- Considering an example of a worker getting injured while working in an assembly line, prepare a sheet for different costs involved and total cost of accident.

## 15.6 LET US SUM UP

The field of industrial accidents and safety analysis is a problematic field of analysis. It is not just linked with the injury alone but also linked to various other factors such as moral, ethical, legal and economic aspects. This chapter focuses on the cost of industrial accident, what are the various types of costs involved in the accident and how to calculate them. With the increase in awareness throughout the world regarding health and safety issues in the industries and work place, the focus on the costs analysis of accidents has also increased. This unit comprises both the technical and organisational aspects to measure accident costs. The technical side of the accident cost analysis is associated with reporting of accidents, accounting of the accidents and controlling of the management so that data acquisition and analysis can be facilitated. Thus, these activities should be linked and integrated with cost

analysis tools. The other side is the organisational side, which deals with the end use and implementation of the accident cost analysis tools.

Comprehensive planning is required in order to promote the sharing and acceptance of the tool within the organisation. Thus, planning and implementation of tools measuring the safety and health issues should be introduced in any organization very carefully and gradually. It must be remembered that, cost analysis of accident does not reflect to achieve zero accidents but it helps management to better understand the need of safety and health at the workplace and serves as a complementary technology to be part of safety culture.

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## 15.7 KEY WORDS

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**Accident:** An accident is an event that causes unintentional damage or injury.

**Incident:** An incident (near-accident) is an event that almost causes unintentional damage or injury.

**Hazard:** The term hazard is often used to denote a possible source or cause of an accident.

**Risk:** It can be defined as the possibility of an undesired consequence.

**Accident investigation:** An accident investigation is the collection and examination of facts related to a specific occurred accident.

**Risk management:** Risk management consists in coordinated activities to direct and control an organization with regard to risk.

**Safety management:** Safety management is a way of managing hazards and risks in an organization.

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## 15.8 ANSWERS TO SAQs

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For all SAQs, please refer to the relevant text in the unit.

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## 15.9 REFERENCES AND FURTHER STUDIES

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## UNIT 16 PREVENTION OF ACCIDENTS

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### Structure

- 16.1 Introduction
  - Objectives
- 16.2 Need for Accident Prevention
- 16.3 Principles of Accident Prevention
- 16.4 Human Factors in Accidents and Accident Prevention
- 16.5 Prerequisites for a Major Hazard Control System
  - 16.5.1 Manpower Requirements
  - 16.5.2 Technical and Safety Equipment
  - 16.5.3 Sources of Information
- 16.6 Analysis of Hazards and Risks
  - 16.6.1 Preliminary Hazard Analysis (PHA)
  - 16.6.2 Hazard and Operability Study (Hazop)
  - 16.6.3 Accident Consequence Analysis
- 16.7 Effective Workplace Inspections for Accident Prevention
- 16.8 Common Practices to Prevent Accidents in the Workplace
- 16.9 Hierarchy of Accident Prevention and Control Measures
  - 16.9.1 Job Safety Analysis
  - 16.9.2 Basic steps to handle emergency in the work place
- 16.10 Good Safety Practices. Case Study: British Sugar (Uk)
- 16.11 Let Us Sum Up
- 16.12 Key Words
- 16.13 Answers to SAQs
- 16.14 Reference and Future Studies

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### 16.1 INTRODUCTION

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Accidents do happen. But, at the same time they can be prevented. Most of the accidents, whether small or large can be prevented. Even if they are not preventable, much can be done to reduce the level of accident consequences by taking the safety measures and preventive actions. Seriousness of the major accident can be reduced to a large scale by informing the potential victims, taking proactive measures and providing the safety equipment's. Serious consequences from major accidents can only be controlled by strictly implementing the safeguards and safety standards, and installing industrial units away from the densely populated areas.

All-inclusive management of on-site programs for prevention of an industrial accident and attentiveness of the involved people depends upon two primary keys:

1. The organized and efficient coordination of technical, administrative, legal and infrastructure considerations and,
2. To realize that the efficient public communication is an effective key to the preparation for the prevention of any occupational accident

### **Objectives**

After studying this unit, you should be able to

- describe the need for prevention of accidents,
- discuss the humanitarian factors associated in accidents and its prevention,
- relate the prerequisites for a major hazard control system,
- analyze the hazard and risk
- carry out effective workplace inspections for accident prevention
- discuss the common practices to be followed to prevent accidents in the workplace.

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## **16.2 NEED FOR ACCIDENT PREVENTION**

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Accident prevention is highly essential in any company or organization. Accident in an industry leads to various losses and adverse effects to the organization apart from direct effects of injury to the employee. Some of the industrial accidents in the past even lead to the complete closure of the plant and legal actions against them. Figure 16.1 shows the different reasons for accident prevention.



**Figure 16.1: Reasons for Accident Prevention**

Accident prevention is highly essential in an industry for the following reasons:

- i) Prevent economic losses: prevention of accident leads to reduction in various costs which otherwise would have to be paid by the company in terms of different direct and indirect costs.
- ii) Improve social values: An organization effectively involved in safety measures and accident preventions at workplace is always regarded as the good organization and thus built the social status of the company.
- iii) Avoid legal actions: In most of the countries it has now become mandatory to adhere to the safety standards in order to prevent the accidents. Not meeting the safety measures may lead to legal actions against the company.
- iv) Avoid loss of productivity: Accident in an industry leads to huge losses to the organization in terms of loss of productivity. Any accident be small or large, always accounts for downtime loss, loss of labour and various other costs which results in reduction in productivity.
- v) Humanitarian ground: Above all, prevention of accident in any industry is a true humanitarian concern. It leads to have good relations between employer and the employee and also boosts the morale of the employees and builds the mutual trust between the employer and the employee.

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### **16.3 PRINCIPLES OF ACCIDENT PREVENTION**

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The core of accident prevention principle consists of foreseeing the unsafe conditions and of rectifying them before accident occurs. Investigation of an accident, even if it has not resulted in injury, aids in pinpointing the deficiencies in the set-up and helps in preventing recurrence of accidents.

- i) Accident prevention is an essential part of good management and of good workmanship.
- ii) Management and workers must co-operate whole heartedly in securing freedom from accidents.
- iii) Top management must take the lead in organizing safety in the works.
- iv) There must be a definite and known safety policy in each work place.
- v) The organization and resources necessarily must exist to carry out the policy.
- vi) The best available knowledge and methods must be applied

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### **16.4 HUMAN FACTORS INVOLVED IN OCCUPATIONAL ACCIDENT AND ITS PREVENTION**

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Human factors play an important role in the accidents as well as in the prevention of accidents as follows:

- i) The model describing causation of accident should take into account relative timing as well as the relation between the elements.
- ii) The factors which leads to an accident varies with respect to their contribution in the accident. The chronological sequence of these factors also plays a key role in deciding the importance of any particular factor responsible for the accident. These two dimensions can vary independently. For example, some factors are important because of their occurrence in a very close time proximity of an accident and thus it can indicate the approximate time of an accident. Moreover, these factors may also be primarily responsible for the accident, or both. The complete analysis of both the chronological as well as causal importance of factors involved in the accident, the true reasons of the occurrence of an accident can be predicted instead of just describing how it occurred.
- iii) The industrial accidents are generally caused by the simultaneous activation of various factors instead of just one or two. In some cases, the poor or faulty interaction between the human, technical and environmental components can be critical and may be the root cause of an accident. However, the established frameworks for the analysis of an accident are restricted by the range of predefined categories. This flaw in the framework leads to the limited amount of information and in turn also restricts the range of options which are predominantly visible for the preventive action. In case of in-depth analysis for the cause of an accident, the large number of factors are explored and consequently, the model has to take an extensive range of factors into consideration. In this wide range, the interaction among the engagement of a human factor with other human and non-human factors are taken into consideration. Similarly, the pattern of occurrences, co-occurrences and inter-relationship of various different elements within the causal network of an accident are analyzed. These information provides a clear understanding of the root cause of an accident.
- iv) The interaction of two considerations such as the nature of an event and nature of its contribution in the accident. The accidents are not always solely occurred by any particular cause. In fact, there may be multiple causes responsible for an accident. However, the role played by these causes in the occurrence of an accident are not equivalent. Thus, understanding of the role played by each factor in the accident becomes critically important to analyze the actual reason for its occurrence and consequently, to prevent the future accidents. For example, the change in the environmental conditions during the accident which should not have occurred in the environmental conditions at the time of establishing the standard procedure. Thus, the prevention would be much more effective if it is targeted towards the hidden underlying causes instead of the immediate cause of an accident. This study of actual causal network can only be done if all type of factors are given considerable weightage in the

study, examination of their relative timing is conducted and their relative importance is predicted.

- v) There can be infinite potential ways in which the human factor could have been directly engaged in the occurrence of an accident. However, out of these infinite possibilities, there are relatively few patterns which accounts for majority of the accidents. In a study conducted by Feyer and Williamson (1991) over 3 years, it was reported that around two-third of occupational accidents in Australia are caused by only four pattern of predominant factors. Interestingly, in almost all of these patterns human factors are involved at some point of accident.

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## 16.5 PREREQUISITES FOR HAZARD CONTROL SYSTEM

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Many operations in the plants are hazardous. Therefore, a well-established safety mechanism and various facilities are pre requisite for operations of such plants.

In general, the working of a major hazard control system essentially requires the manpower, advanced equipment's and last but importantly the information source. These essential requirements are discussed in detail as follows:

### 16.5.1 Manpower Requirements

Any organization going to construct or working on any major hazard related installations should ensure availability of adequate number of workers with sufficient expertise in the field. Prolonged hours of working may also lead to the risk of accident seen by the experts. Thus, the design of jobs and systems of working hours should be systematically arranged thus reducing the pressure of work and the risk of accidents.

The competent and responsible authorities frames the rules and regulations for the fully operational major occupational hazard control system. These regulations provides the mandatory guidelines for the engagement of following specialized and skilled manpower:

- a) Government officials (inspectors, commissioners etc.) with special powers;
- b) Hazard and risk assessment professionals;
- c) Qualified personnel for conducting examinations and testing's on pressure vessels;
- d) Skilled professional to handle the emergency situations and its management;
- e) Land-use planning experts;
- f) Emergency services such as fire fighters, police authorities, medical professionals etc.

### **Government officials**

The competent authorities are responsible for providing the suitable staff including the qualified professionals for testing and inspection of the hazard control system. Moreover, it also provide specialized training to the officials and conduct frequent mock drills to evaluate the performance of the people involved in hazard control system.

### **Group of Experts**

The group of experts involves engineers and scientists from different fields who work together with an objective to minimize the chances of an occupational accident. However, it is the responsibility of the competent authorities to establish a group of experts in the country and to also provide them with specialized resources for their efficient functioning.

### **Advisory committee**

Advisory committee is framed by the competent authorities to obtain valuable suggestions for the efficient working of the hazard control system. This committee comprises of people from all the involved organizations and also the representatives which are experienced in handling the major hazards such as:

- a) Regulating authorities;
- b) Works managements and employers' organizations;
- c) Employee or trade unions representatives;
- d) Local governing authorities;
- e) Scientific labs and institutions.

The advisory committee is framed with following key objectives:

- a) To discuss the priorities of the major occupational hazard control system with any national requirements.
- b) To raise the technical difficulties in the implementation and efficient working of hazard control system.
- c) To suggest necessary recommendations and instructions on all safety features of hazard control system.

### **16.5.2 Technical and Safety Equipment**

Competent authorities takes the decision of making the data base of national and state inventories of major hazard control systems. This element enables the quick support system from other facilities in case of an accident and the safety equipment's can be made available at the site as early as possible.

The works management and local authorities are responsible for making the availability of technical and safety equipment's in case of an emergency situation. However, this availability of resources varies with the needs of emergency plans. These equipment's includes:

- a) First-aid medications and rescue equipment's;
- b) Fire-fighting equipment's.
- c) Spill control equipment's;
- d) Personal safety devices for rescue staffs;
- e) Measuring devices for various toxic substances;
- f) Antidotes for curing of persons affected by toxic substances.

### **16.5.3 Sources of Information**

The regulating authorities should identify their information requirements for the establishing and efficient working of the major hazard control system.

These requirements may include:

- a) Latest technological developments and advances in the process industries;
- b) Recent progresses made in the major hazard control systems;
- c) Practice codes for the health and safety related technical problems;
- d) Reports of the accidents, its complete evaluation study and remarks;
- e) Critical information's and reports provided by experts and professionals on major hazard control.

The regulating authorities can obtain these information's from various sources such as:

- a) Industry professionals and scientists;
- b) Industry and trade groups;
- c) Organizations which sets the national and international standards;
- d) Employee union;
- e) Scientific institutes and research labs;
- f) International codes of practice and guiding principles;
- g) National codes and guidelines;
- h) Previous accident reports;
- i) Research articles about major hazard valuations;
- j) Conference proceedings;
- k) National and international seminars

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## 16.6 HAZARDS AND RISKS ANALYSIS

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Primarily, works management is responsible for various hazard analysis. However, the same analysis methodology can also be adopted by the competent authorities for the assessment of the safety system.

A hazard analysis should cover the wide areas for the analysis of the safety of a major hazard and its prospective hazards. These wide areas includes:

- a) Identification of hazardous toxic, reactive, explosive and flammable materials in the installation;
- b) Identification of possible failure and errors which may lead to the abnormal conditions and contribute in the major accident;
- c) Identification of the consequences for the workers, people living or working outside the installation etc. in case of major accident;
- d) Prevention system and measures for major and minor accidents;
- e) Mitigation of the accident's consequences.

A formalized method is normally followed for the risk and hazard analysis. This method ensures the consideration of broad areas and its comparability for the efficient working of the prevention system.

### 16.6.1 Preliminary Hazard Analysis (PHA)

- i) PHA is the initial step for any hazard analysis.
- ii) PHA identifies the potential causes which can lead to an accident in the installation. This includes release of toxic fluids, explosive or flammable substances in the installation. The PHA also ensures the basic elements of safety in the system.
- iii) The PHA maintains the documentation of each and every accident, the involvement of components such as (pressure vessels, storage vessels etc.) in the accident, the events or activities which are responsible for the initiation of an accident, and lastly the corresponding safety devices such as safety valves, pressure gauges etc.
- iv) The findings of PHA should comprise the critical information about the units in the installation which requires additional attention and detailed examination. Similarly, the units which are less significant from a major hazard point of view should be also be identified.

### 16.6.2 Hazard and Operability Study (Hazop)

- i) A HAZOP is designed to determine the nonconformities from the normal working conditions in the installation. The HAZOP study also identifies the operational malfunction which may result in the uncontrolled conditions and consequently major accident.

- ii) The HAZOP study is very much needed during the design stage of the new unit's installation. Similarly, HAZOP study is also essentially required if some modifications (due to operational or legal reasons) are to be made in the existing installation.
- iii) The HAZOP study documents the systematic questionnaire regarding every critical part of the design, the motivation or intention of design, its deviation from the intention and potential hazardous conditions.
- iv) A HAZOP study should be necessarily performed by an expert group of people from multidisciplinary backgrounds. Moreover, it should always include the workers or staff who are familiar with the installation.
- v) The HAZOP study group should be headed by an experienced specialist from works management and should have undergone special training from the registered consultant.
- vi) The HAZOP study group should be headed by an experienced specialist from works management or an official specially trained by the professional consultant.

### 16.6.3 Accident Consequence Analysis

The last stage of hazard analysis consists of the investigation on the consequences of an accident. This analysis predicts the consequences of a major accident in the installation on workers, neighborhood and environment.

- i) An accident consequence examination should consist of:
  - a) Details of the accident such as tank rupture, safety valve failure etc.;
  - b) Estimated quantity of toxic and flammable material released due to accident;
  - c) Approximate calculation of dispersion of released material (gas or evaporating liquid);
  - d) Assessment of the dangerous effects such as toxic gases, heat radiation etc.
- ii) The accident consequence analysis methodology should include prototypes for dispersion of pollutant in the atmosphere, thermal radiation etc. The type of physical models varies from the installation to installation depending upon the hazardous substances present in the installation.
- iii) The findings of consequence analysis should be used to determine the protective measures needed in the installation. This may include fire-fighting system, alarms, pressure relief systems etc.

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## 16.7 EFFECTIVE WORKPLACE INSPECTIONS FOR ACCIDENT PREVENTION

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Workplace inspections help prevent incidents, injuries and illnesses. Through a critical examination of the workplace, inspections help to identify and record hazards for corrective action. Health and safety committees can help plan, conduct, report and monitor inspections. Regular workplace inspections are an important part of the overall occupational health and safety program and management system, if present.

Inspections are important as they allow you to:

- listen to the concerns of workers and supervisors
- gain further understanding of jobs and tasks
- identify existing and potential hazards
- determine underlying causes of hazards
- recommend corrective action
- monitor steps taken to eliminate hazards or control the risk (e.g., engineering controls, administrative controls, policies, procedures, personal protective equipment)

Planning is essential for an effective inspection.

### **What to Examine**

The inspection should be focused on who, what, where, when and how. During inspection, an additional attention is given to the items that are potential source for the unsafe and unhealthy conditions owing to the stress, wear, impact, vibration etc. Inspection should also include the areas where work is not done on the regular basis such as parking lots, rest areas and locker rooms.

### **Workplace related elements**

The workplace elements comprises of the people, environment, equipment's and the process.

The environment indicates the hazards through noise, vibration, lighting and temperature. Equipment includes the involved raw materials, tools, jigs and fixtures, for the manufacturing of a product. The way of interaction between the workers and other elements in a series of operations comes under the process element.

### **What types of hazards do we look for in a workplace?**

Various types of workplace related hazards are:

- Safety hazards includes the threats triggered by the poor machine guards, risky workplace circumstances and unsafe work practices.

- The hazards caused by the micro-organisms such as bacteria, viruses etc. comes under the category of biological hazards.
- The threats associated with solid, liquid, gases, dust etc. belongs to the chemical hazards.
- Physiological and psychological pressure on the workers due to various activities such as repetitive and forced movements, incompatible work postures, improper designed workplace etc. leads to the ergonomic hazards.
- The condition of the work atmosphere such as noise, vibration, heat, radiation etc. comes under the category of physical hazards.
- The various other mental conditions of the worker due to overwork, stress, bullying etc. are associated with the psychological hazards.

Information required for the inspection report includes the following:

### **Area diagram**

The plant layout and floor plans assist in the planning of an efficient inspection in the installation. The entire installation can be sub-divided into various zones based on the processes. The activities in these sub-divided zones are then analyzed visually. Moreover, an additional attention is invested on the location of machinery, equipment and raw as well as finished materials in the zone. Material and work flow in the installation is critically analyzed with a special attention to air ducts, aisles, stairways and fire exits. Opinion of workers and supervisors are also given due consideration as they are more familiar to the work place.

### **Equipment Inventory**

Information on the inventory i.e. machinery, equipment, raw materials, finished goods etc. also contribute significantly in planning of an efficient inspection methodology. Besides inventory information, the technical data sheets and manufacturer safety manuals should also be given due consideration during inspection. Work area records provides the critical information related to the hazards associated with the equipment and thus it should also be given significant weightage during the inspection.

### **Hazardous Product or Chemical Inventory**

The products engaged in the production should be listed along with its data sheets availability information. A check should be made on the exposure sources. The training and information on how to use, store and handle the products is to be necessarily imparted to the workers. The hazardous products should be properly labelled as per the Workplace Hazardous Materials Information System (WHMIS) guidelines.

## Checklists

The checklist assists in the inspection responsibilities, controlling the inspection activities, and lastly documentation and preparation of inspection reports. Checklist also help in taking the on-spot readings and comments. However, while maintaining the checklist, there are chances of missing some other hazardous conditions which should be avoided. Thus, the checklists should only be used as a basic tool. A customized checklist as shown in Figure 16.2 is often more appropriate than the consolidated one.

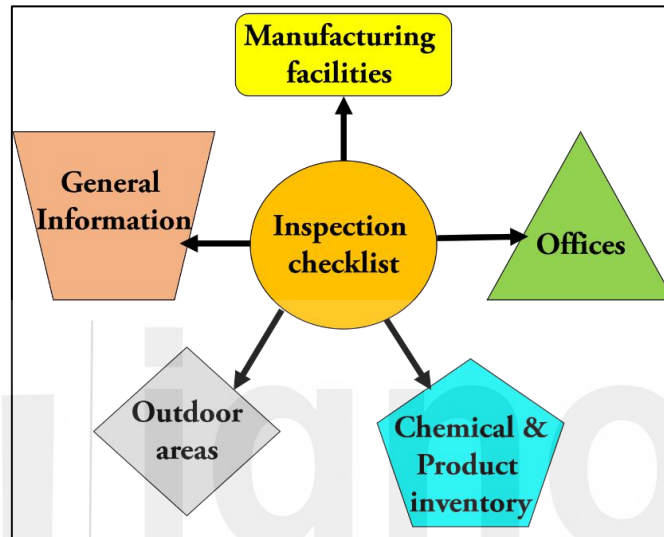


Figure 16.2: Customized Check List for Inspection

## Reports

Inspection report is a crucial document and should be maintained properly. Past inspection report indicates the flaws which were previously identified and set a road map for the future inspections. It also provides the information regarding the areas which were not included in the previous inspections. The inspection report should not be copied from the previous reports, instead more attention should be paid on the critical elements which were mentioned in the previous reports. The past reports can be used to look for the issues and a check should be made whether the recommendations are implemented or not. The effective changes should be recorded down.

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## 16.8 COMMON PRACTICES TO PREVENT ACCIDENTS IN THE WORKPLACE

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- i) **Create a Safe Work Culture:** The workplace will be safe only when it is on the first priority of both the employers as well as of employees. Unfortunately, the safe work culture is not promoted by all the employers and as result of it a careless attitude is developed in the workplace. The employers should reward or recognize the workers who are practicing the safe work habits without hampering the production rate. These type of initiatives encourage workers to perform in a better way

and such positive steps also significantly contribute for the long run and safe workplace culture.

- ii) **Implement Fall Prevention Systems:** The occupational safety and health administration (OSHA) drafts the fall prevention regulations which is to be necessarily followed by all the organizations. The OSHA standards compels the manufacturer to find the fall prevention equipment as per the particular need.
- iii) **Perform Preventative Maintenance:** The severe accidents and injuries can be significantly reduced by conducting the preventive maintenance of equipment's in the installation. In addition to the safe work culture, preventive maintenance also extends the life span of equipment's involved in the production. The potential hazards can be identified and effectively cured in the regular preventive maintenance and thereby avoids the probability of accidents.
- iv) **Keep the Workplace Clean:** The clean work area is essential for the effective production rates. However, in manufacturing industries the cleaning of work sites is rather difficult but it can be a disaster for companies. The management should provide guidelines to the workers for keeping their work area clean which can also reduce the risk of falls and trips. The awareness programs for workers can be organized by managements for keeping the work area clean for the workers own safety.
- v) **Provide safety training for all employees:** The employees should be aware and prepared for the work hazard that may arise due to their nature of job. This is the responsibility of management to provide adequate training on all the work place safety standards and hazards to the workers. Workers should also provide a genuine feedback about the health and safety policies for their job. The documented safety policy should include the first-aid procedure for the each and every injury that may cause in the industry. It should also mention the name and location of the trained first-aid personnel. The employees should not operate the equipment's on which they are not having proper training.
- vi) **Utilize protective clothing and gear:** Many jobs require a specially designed uniform as per its nature of interaction with the workplace. The workers should wear proper uniforms and other recommended safety equipment's during working. These equipments may include hard hat, high-visibility clothing, goggles, protective suits etc.
- vii) **Avoiding Safety Shortcuts:** The shortcuts at the cost of safety are never recommended in the work place. It is always better to follow the standard procedure for the safe work environment of own and fellow workers in the organization.
- viii) **Miscellaneous : Some other practices are**
  - a) Lifting Quickly
  - b) Skipping Breaks
  - c) Using Tools Improperly
  - d) Applying the Wrong Safety Label

- e) Stock Ignoring 'Near Misses'
- f) Insufficient Emergency Equipment
- g) No Clear Emergency Exit Path

By investigating every incident, we learn about causes and can take actions towards mitigating or removing the causes.

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## **16.9 HEIRARCHY OF ACCIDENT PREVENTION AND CONTROL MEASURES**

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An accident is an unexpected, unplanned event in a sequence of events that occurs through a combination of causes which results in physical harm, injury or disease to an individual, damage to property, a near miss, a loss or any combination of these effects. Risks should be avoided / eliminated and if not possible reduced by taking preventive measures. The order of priority is also known as the hierarchy of control. There are five common steps in the heirarchy of accident control.

### **Step-1: Elimination**

Elimination of hazards refers to the total removal of the hazards. Elimination means that a risk is reduced to zero. Elimination is the ideal objective of any risk management. This is a permanent solution and should be attempted in the first instance. If the hazard is removed, all other management controls, such as work place monitoring, training, safety, auditing and record keeping will no longer be required.

### **Step-2: Substitution**

Substitution means replacing the hazard by one that presents a lower risk. The concept of replacing the dangerous chemicals by the non-dangerous or the less dangerous chemicals can be applied. Step-3: Engineering control: Engineering controls are physical means that limit the hazard. These include structural changes to the work environment or work processes. Local exhaust ventilation (LEV) to control risks from dust or fume is a common example. Separation of the hazard from operators by methods such as enclosing or guarding items of machinery and equipment.

### **Step-4: Administrative controls**

Administrative controls are also known as organizational measures. Administrative controls reduce or eliminate exposure to a hazard by adherence to procedures or instructions. Documentation should emphasise all the steps to be taken and the controls to be used in carrying out the activity safely.

### **Step-5: Personal protective equipment (PPE)**

Personal protective equipment (PPE) should be used only as a last resort. Thus accident prevention has been based on learning from accidents and near misses incidents. By investigating every incident, we learn about causes and can take actions towards mitigating or removing the causes.

### **16.8.1 Job Safety Analysis (JSA)**

Job Safety Analysis (JSA) is a procedure that identifies the hazard or potential accident associated with each step of a job. It helps to develop

solutions that will either eliminate or guard against hazards. Job safety analysis is a procedure which helps in integrating accepted safety and health principles into a particular task or job operation. In job safety analysis, each basic step of the job is to identify potential hazards and to recommend the safest way to do the job. There should be a group of experienced workers and supervisors in order to complete the analysis through discussion. An advantage of this method is that more people are involved in a wider base of experience and promoting a more ready acceptance of the resulting work procedures. Basic steps for making a job safety analysis There are four basic steps for making a job safety analysis:

- a) Select the job to be analyzed
- b) Breakdown the jobs into successive steps
- c) Identify the Hazards in each step and the potential of Accidents
- d) Develop ways to eliminate hazards and potential accidents. Hence job safety analysis produces safe job procedures that guide a worker through a task from start to finish orderly and safely which in turn serves as a best tool in accident prevention program.

### **16.8.2 Basic steps to Handle Emergencies in the Work Place**

Emergencies include accidental releases of toxic gases, chemical spills, fires, explosions and bodily harm.

#### **Planning:**

The effectiveness of response during emergencies depends on the amount of planning and training performed. Management must show its support for plant safety programs and the importance of emergency planning. If management is not interested in employee protection and in minimizing property loss, little can be done to promote a safe work place. It is therefore management's responsibility to see that a program is instituted and that it is frequently reviewed and updated. The input and support of all employees must be obtained to ensure an effective program. The emergency response plan should be developed locally and should be comprehensive enough to deal with all types of 13 emergencies specific to that site. The plan must be in writing and it must include as a minimum the following elements:

- Emergency escape procedures and emergency escape route assignments. Procedures to be followed by employees who remain to perform or shut down critical plant operations before the plant is evacuated.
- Procedures to account for all employees after emergency evacuation have been completed.
- Rescue and medical duties for those employees who are to perform them.
- The preferred means for reporting fires and other emergencies.
- Names of regular job titles of persons or departments to be contacted for further information or explanation of duties under the plan.

The emergency action plan should address all potential emergencies that can be expected in the work place. Therefore it will be necessary to perform a hazard audit to determine toxic materials in the work place, hazards and potentially dangerous conditions. For information on chemicals, the manufacturer or supplier can be contacted to obtain material safety data sheets. These forms describe the hazards that a chemical may present, list precautions to take when handling, storing or using the substance and outline emergency and first-aid procedures. The employer must list in detail the procedures to be taken by those employees who must remain behind care for essential plant operations until their evacuation becomes absolutely necessary. This may include monitoring plant power supplies, water supplies and other essential services that can not be shut down for every emergency alarm and use of fire extinguishers. For emergency evacuation, the use of floor plans or work place maps that clearly show that the emergency escape routes and safe or refuge areas should be included in the plan. All employees must be told what actions they are to take in emergency situations that may occur in the work place such as a designated meeting location after evacuation. This plan must be reviewed with employees initially when the plan is developed, whenever the employees responsibilities under the plan change and whenever the plan is changed. A copy should be kept where employees can refer to it at convenient times. In fact, to go a step further, the employer could provide the employees with a copy of the plan, particularly all new employees.

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## **16.10 GOOD SAFETY PRACTICES. CASE STUDY: BRITISH SUGAR (UK)**

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### **Background:**

In 2003, two fatal accidents occurred at British sugar despite of having a well-equipped occupational health and safety provisions. These accidents caused a casualty of two workers. After the tragedy, the organization invested a heavy fund for the complete transformation of the work procedures and for the better implementation of the safety measures. British sugar's health and safety management model is shown in Figure 16.3.

### **Difficulties**

It was observed, that **the workers have not adapted the new work procedures** even after the huge capital investment made by the organization in the transformation. This work behaviour is due to the **sense of insecurity among the workers** regarding the new safety provisions which were initiated by the organization.

### **Solution**

As a solution to the problem the company came up with a proposal of "Safety debate" program as an implementation tool in 2006. It is an E-learning platform which is developed by the British sugar. This platform facilitates the

confidential discussion between the managers, middle managers and workers regarding the plant and work safety. The objectives of this program were as follows:

- **Mutual anonymous discussion between the managers and workers.**
- Planning cooperative steps for the improvement of safety procedures
- Implementation of new technologies, communication procedures and action plans on paper.

### Benefits

- The accident frequency rate (i.e. number of accidents per million hours) was decreased from 5.2 in financial year 2005-06 to 3 in the financial year 2008-09 making an overall improvement of ~ 40%.
- The ratio of safety observations in 2005-06 was drastically increased by 346% in year 2008-09. The observation of unsafe acts was also increased from the ratio of 4:1 to 20:1 after the implementation of program. These statistical data indicates an excellent commitment and involvement in the safety procedures at all production levels.
- The program have significantly improved the procedures and work standards.
- In 2009, the British Sugar was awarded the “DuPont safety award” for its effective and innovative approach for the safety management.

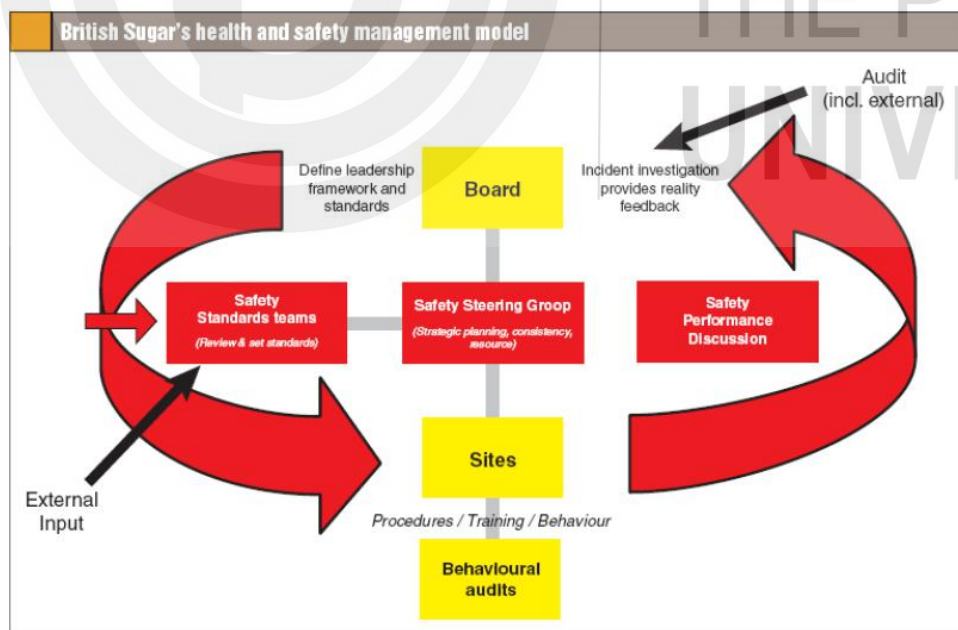


Figure 16.3: British sugar's health and safety management model

### Conclusions

From this case study, it can be concluded that the huge investments are not required in all the cases and instead, an innovative approach with an active

participation of the human capital can serve the purpose in context of health and safety measures in the organization.

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## 16.11 LET US SUM UP

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**How to achieve Safety:** The important pillars to achieve the safety are shown in Figure 16.4.



Figure 16.4: Important pillars to achieve safety

Broadly it is grouped in to 5 Es methods;

- 1) Engineering - Process Control
- 2) Enforcement -Rules & Procedures
- 3) Education - Training
- 4) Enthusiasm -Behavioral Aspect
- 5) Evaluation - Audit, Mock Drills

A trained workforce alert of hazards, aware of guards & facilities and also aware of the need to work safely is indeed an asset because it is ultimately the safety performance on the shop floor that matters. Hence human touch to all your shop floor policies is an important strategy in any Organisation for a Total Disaster Prevention Program.

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## 16.12 KEY WORDS

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**Accident:** An accident is an event that causes unintentional damage or injury.

**Incident:** An incident (near-accident) is an event that almost causes unintentional damage or injury.

**Hazard:** The term hazard is often used to denote a possible source or cause of an accident.

**Risk:** It can be defined as the possibility of an undesired consequence.

**Accident investigation:** An accident investigation is the collection and examination of facts related to a specific occurred accident.

**Risk management:** Risk management consists in coordinated activities to direct and control an organization with regard to risk.

**Safety management:** Safety management is a way of managing hazards and risks in an organization.

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## 16.13 ANSWERS TO SAQs

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For answers to SAQs, please refer to the relevant text in the unit.

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## 16.14 REFERENCES AND FURTHER STUDIES

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**MIS – 021**  
**SAFETY PHILOSOPHY AND PRINCIPLES OF ACCIDENT**  
**PREVENTION**

**BLOCK 1 INTRODUCTION TO INDUSTRIAL SAFETY**

|        |                                     |
|--------|-------------------------------------|
| UNIT 1 | Basic Concept of Industrial Safety  |
| UNIT 2 | Safe Working Practices              |
| UNIT 3 | Personal Protective Equipment (PPE) |
| UNIT 4 | Fire Safety                         |

**BLOCK 2 INTRODUCTION TO INDUSTRIAL SAFETY**  
**ENGINEERING**

|        |                                                            |
|--------|------------------------------------------------------------|
| UNIT 5 | Concept of Safety Engineering (Ergonomics, Process Safety) |
| UNIT 6 | Storage of Material Handling of Hazardous Material         |
| UNIT 7 | House Keeping (5S Concepts)                                |
| UNIT 8 | Safeguarding of Machinery                                  |

**BLOCK 3 INTRODUCTION TO INDUSTRIAL SAFETY**  
**MANAGEMENT**

|         |                                 |
|---------|---------------------------------|
| UNIT 9  | Safety Organizations            |
| UNIT 10 | Safety Policy                   |
| UNIT 11 | Training and Awareness Creation |
| UNIT 12 | Safety Audit                    |

**BLOCK 4 INDUSTRIAL ACCIDENTS AND ITS**  
**PREVENTION**

|         |                                     |
|---------|-------------------------------------|
| UNIT 13 | Introduction to Industrial Accident |
| UNIT 14 | Types of Accidents and Its Analysis |
| UNIT 15 | Cost of Accidents                   |
| UNIT 16 | Prevention of Accidents             |